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Aviation



Airport Local Air Quality Studies with OpenALAQS



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NETWORK
MANAGER



Agenda

☐ Introduction

- ☐ *Key Objectives of the Training Course*
- ☐ *General information about OpenALAQS/QGIS*

☐ OpenALAQS:

- ☐ *Create a New Study*
- ☐ *Define Emission Sources*
- ☐ *Generate Emissions Inventory*
- ☐ *Calculate & visualize emissions*
- ☐ *Useful information about the emissions calculation methods*

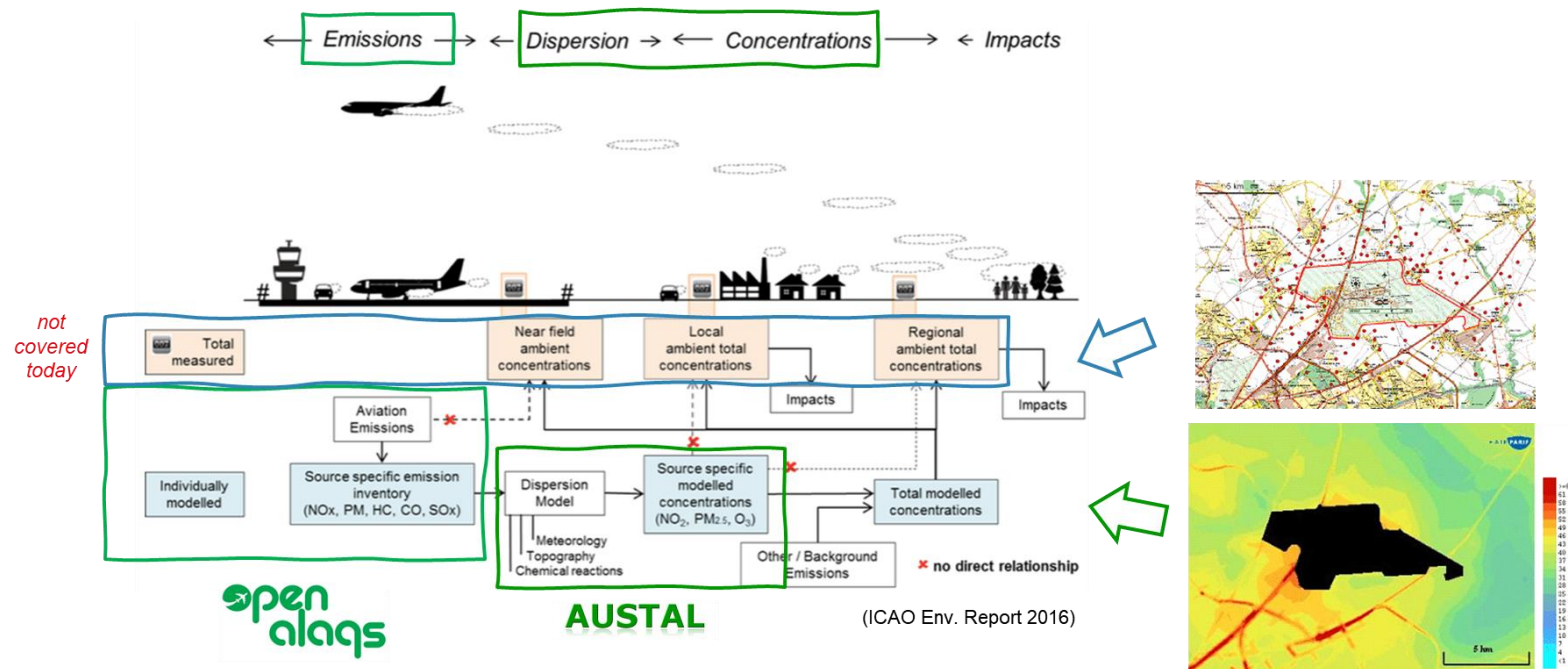
☐ Dispersion Modeling with AUSTAL

- ☐ *Data Preparation*
- ☐ *Calculate & visualize concentrations*

☐ Q&A

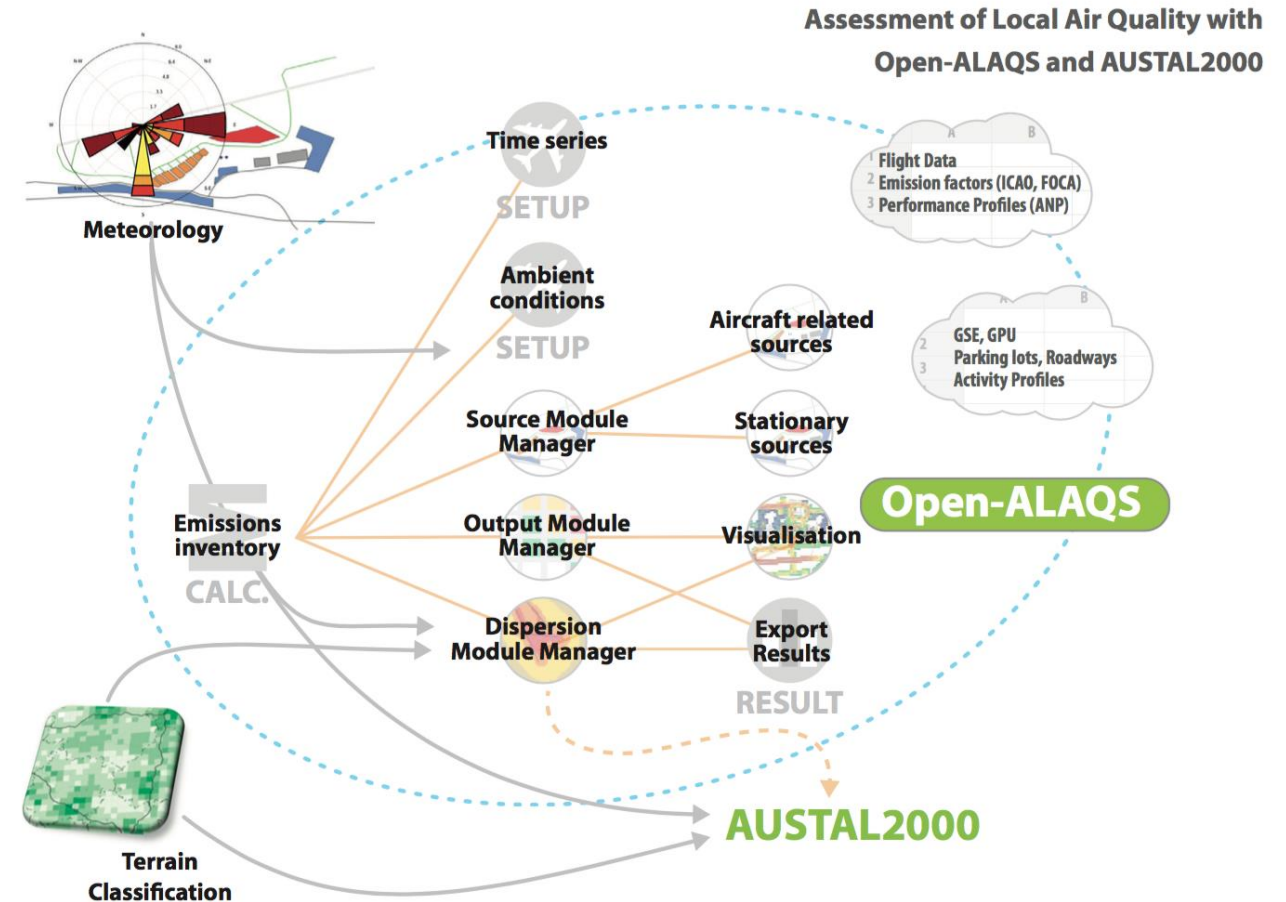
Key Objectives of the Training Course

- ❑ Describe the core features of OpenALAQS for assessing airport emissions
- ❑ Explore the modelling capabilities of OpenALAQS
- ❑ Interpret results and understand their implications for air quality management at airports

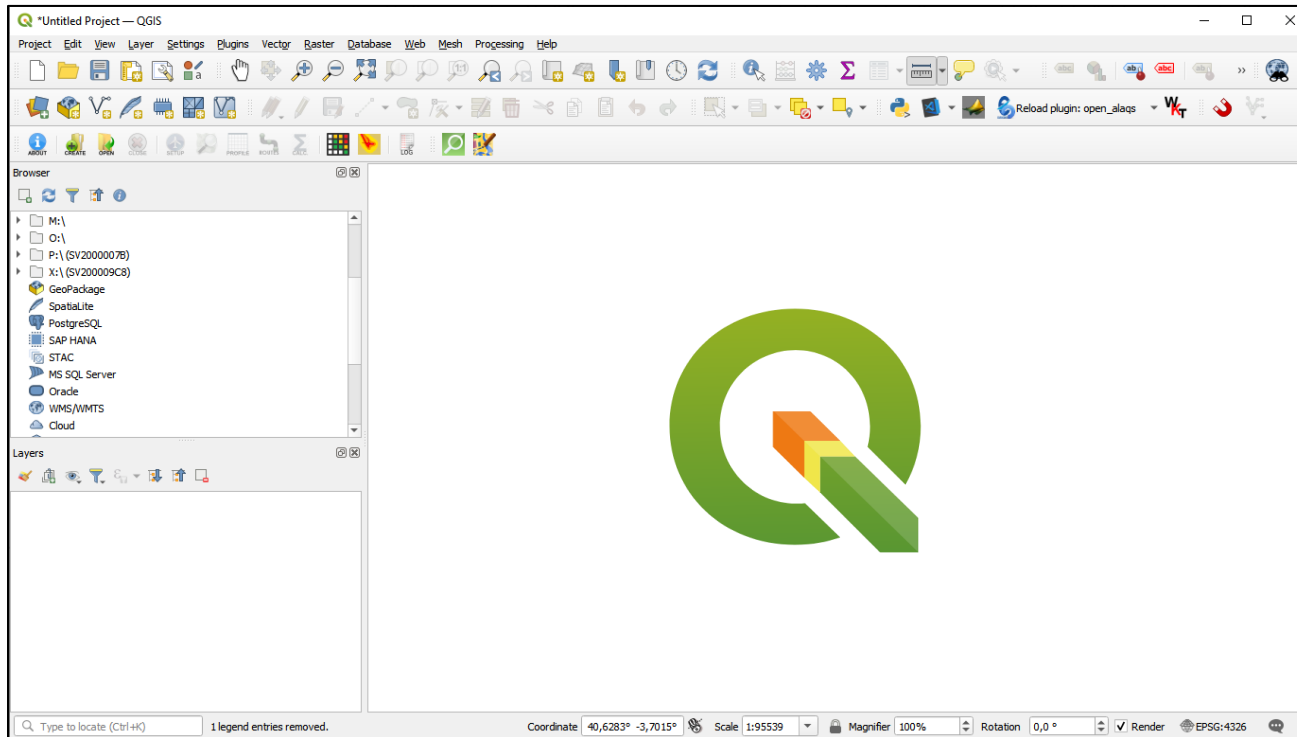


Open-ALAQs

- ❑ Developed as plugin for QGIS
- ❑ Full LAQ modelling suite (with AUSTAL)
- ❑ Modular architecture
- ❑ Code publicly available on GitHub
https://github.com/eurocontrol-asu/open_alaq



Short introduction to QGIS



Archive

[About](#) [Resources](#) [Community](#) [Download](#) [Donate](#)

Resources

**Installation troubles?**

If you are looking for information on installing QGIS on all available platforms, we have prepared a comprehensive guide for you:

[Installation guide](#)

Documentation

QGIS has a lot of documentation. All of the core documentation (user manual, developer handbooks etc.) is available in English. Some documents such as the user guide are also available in other languages.

You will find documentation for every QGIS Long Term Release (LTR) on the respective documentation website.

[Get involved](#) and help us write a better documentation.

Choose your language: English

[QGIS 3.44](#) [QGIS 4.0](#) [QGIS testing \(>4.0\)](#) [Archived releases](#) [Data protection](#)

For users (QGIS 3.44):

[Desktop User Guide — English](#) [QGIS Training manual — English](#) [Gentle Intro to GIS — English](#)

[Server Guide — English](#)

<https://qgis.org/resources/hub/>

Install OpenALAQS

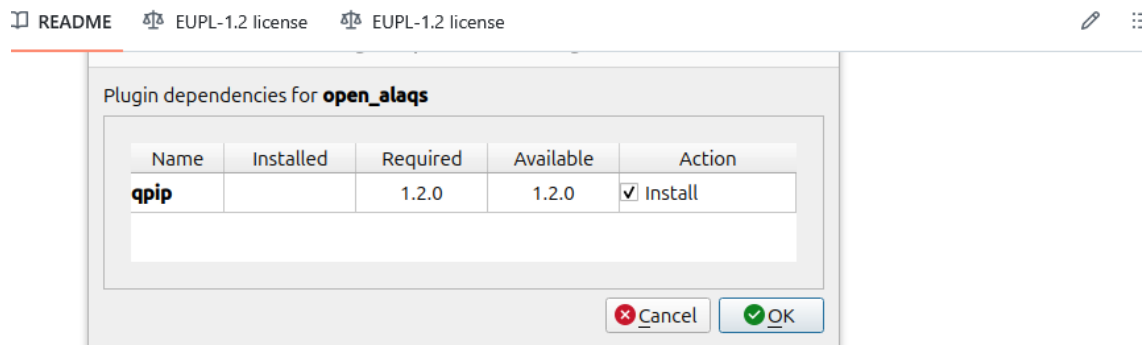
Install OpenALAQS

You can download OpenALAQS [latest release from GitHub](#), or browse [previous releases](#).

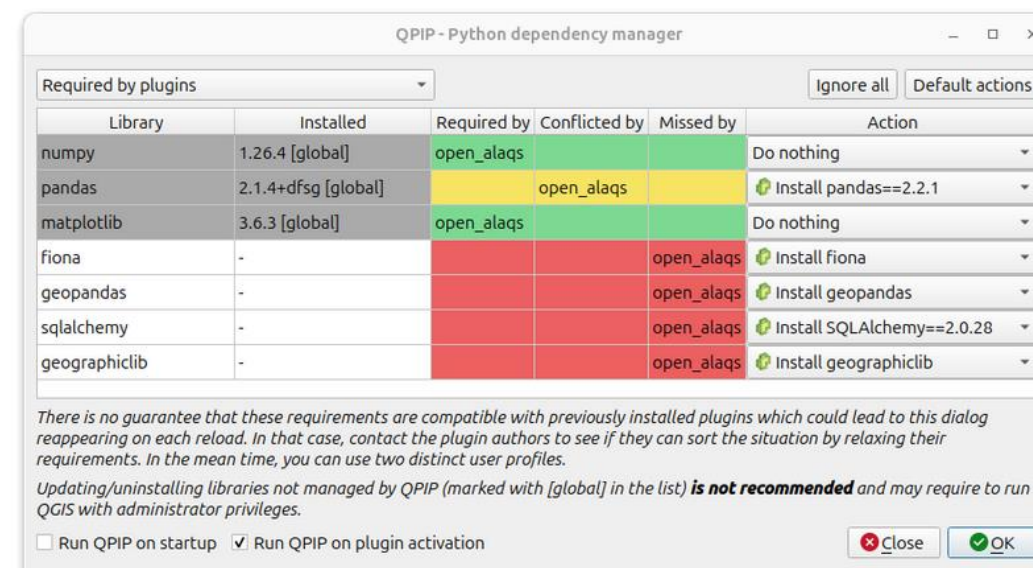
Once you download the .zip file, go to QGIS, open the "Plugins", then "Manage and Install Plugins...". In the newly opened window, select the "Install from ZIP" on the left sidebar. Then select the recently downloaded .zip file and click "Install Plugin".

While installing OpenALAQS:

- QGIS will first ask you to install its plugin dependency, called QPIP. Click **OK**.

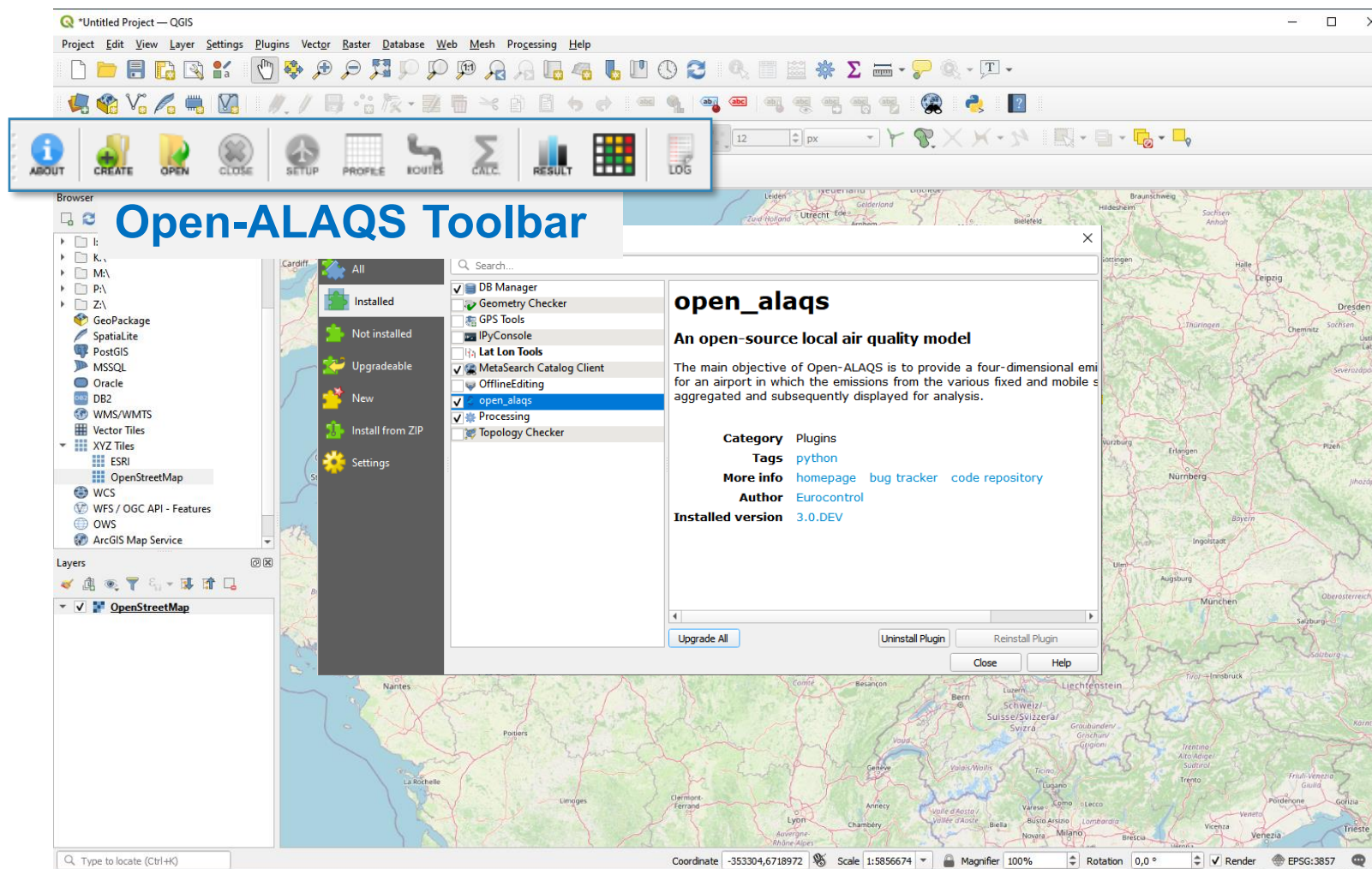


- Once QPIP is installed, QPIP asks to install the Python dependencies of OpenALAQS. If there is any conflicted dependency (marked in yellow), make sure you select a proper **Action** that allows OpenALAQS to find the required dependency version, like in the image below. Then click **OK**.

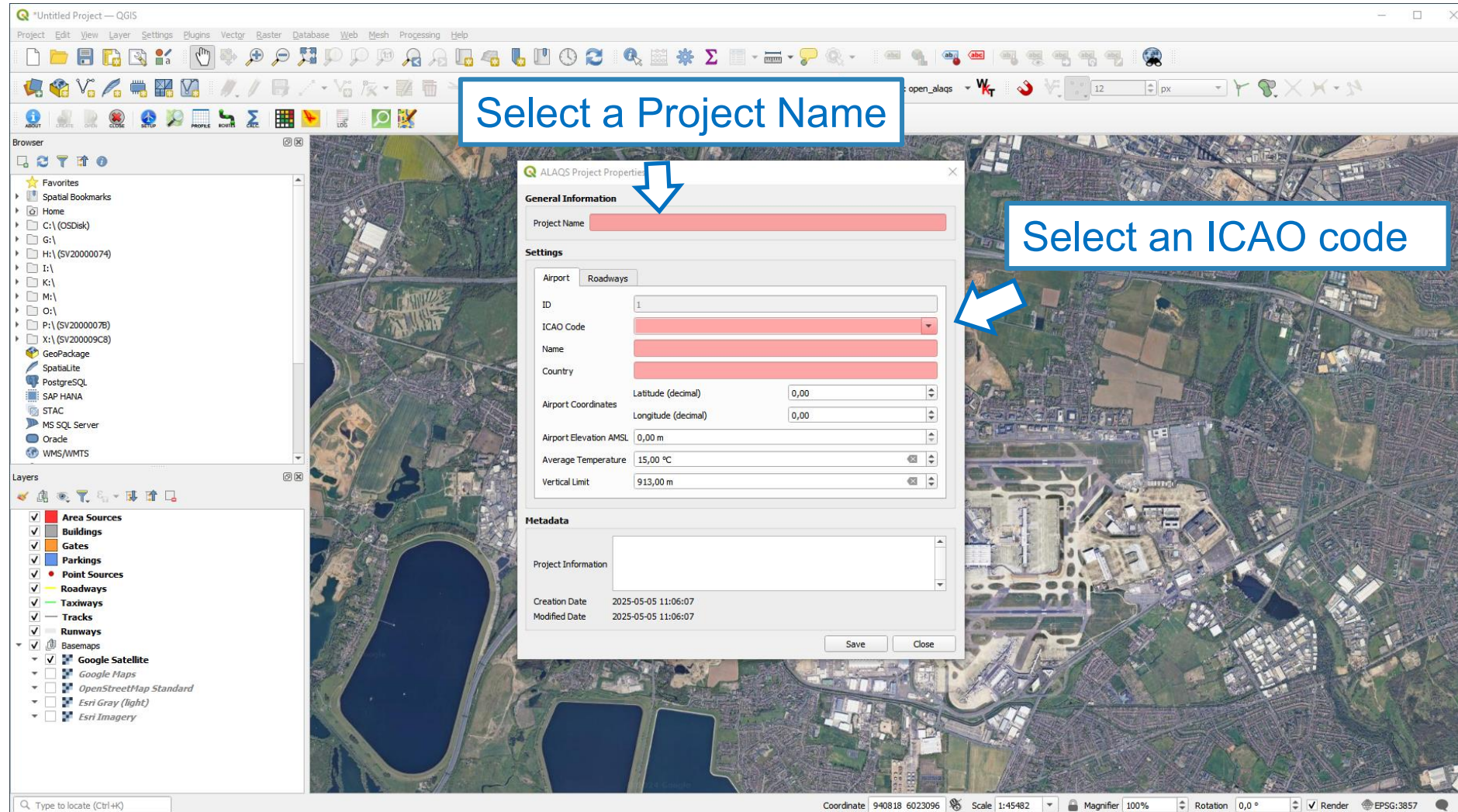


https://github.com/eurocontrol-asu/open_alajs#installation

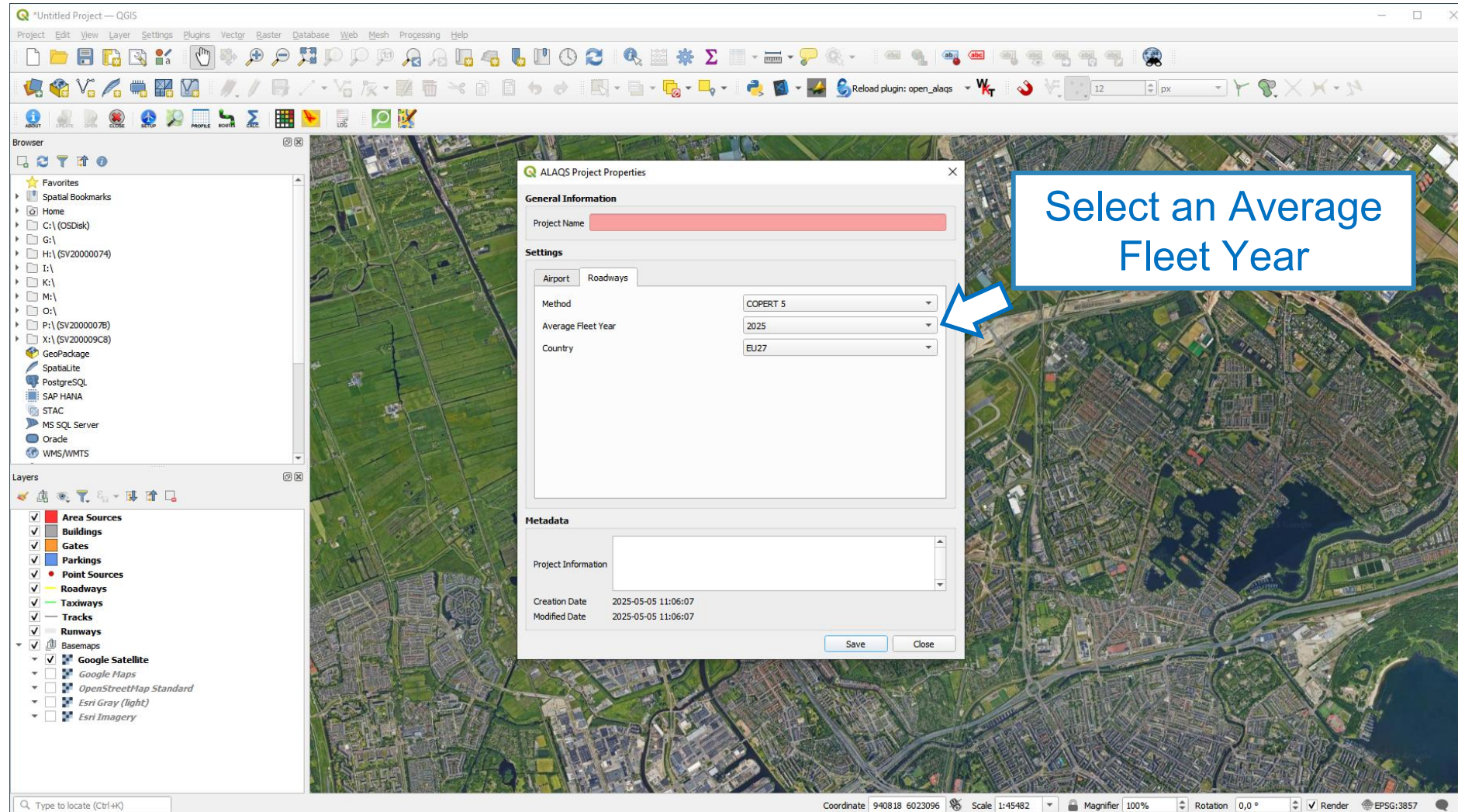
Open-ALAQS / QGIS



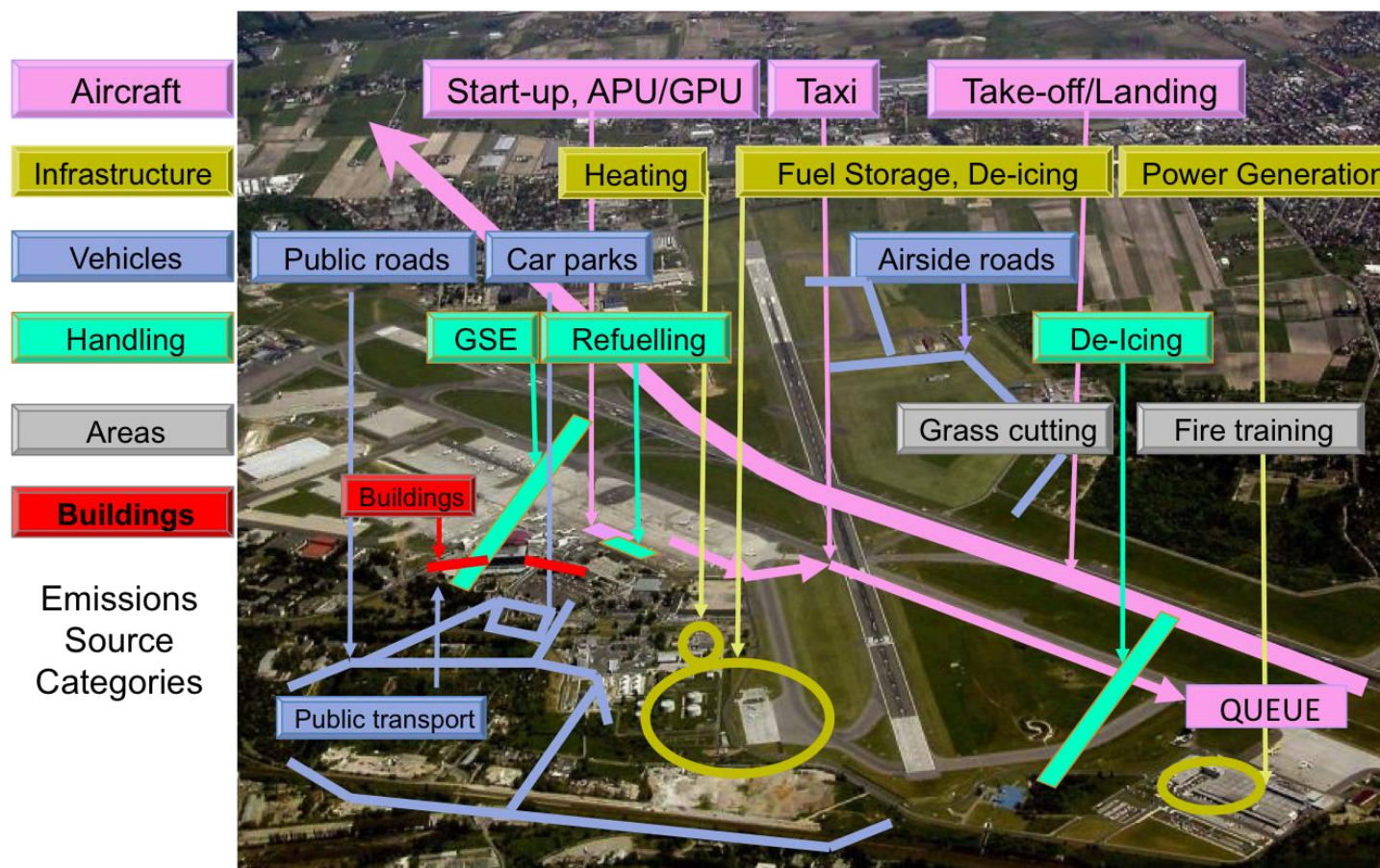
Create a new study



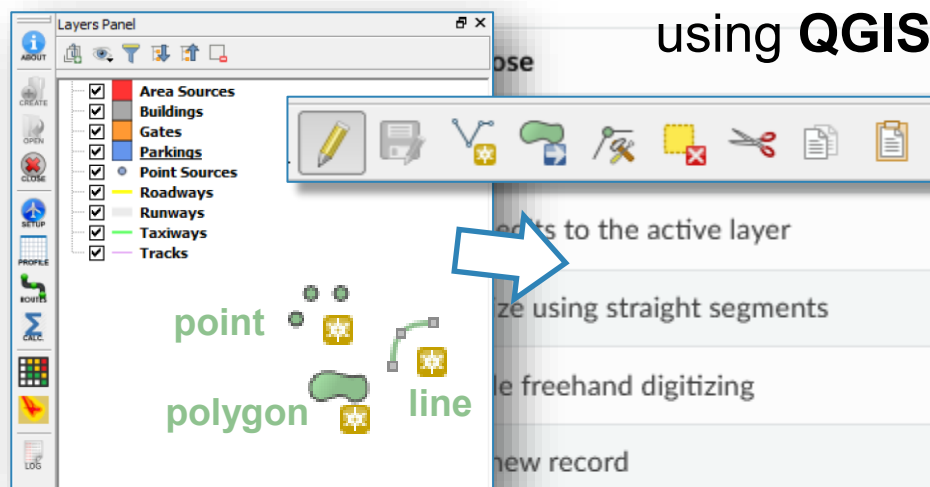
Create a new study



Definition of Emission Sources



Definition of Emission Sources

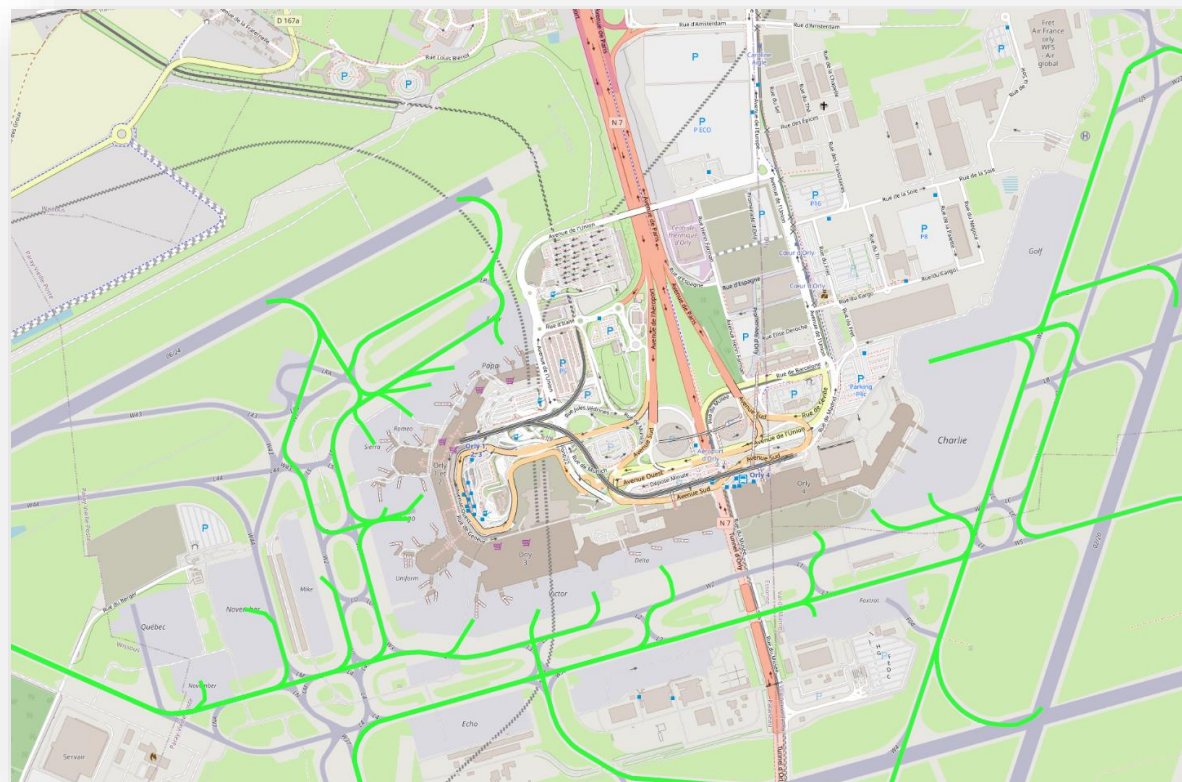
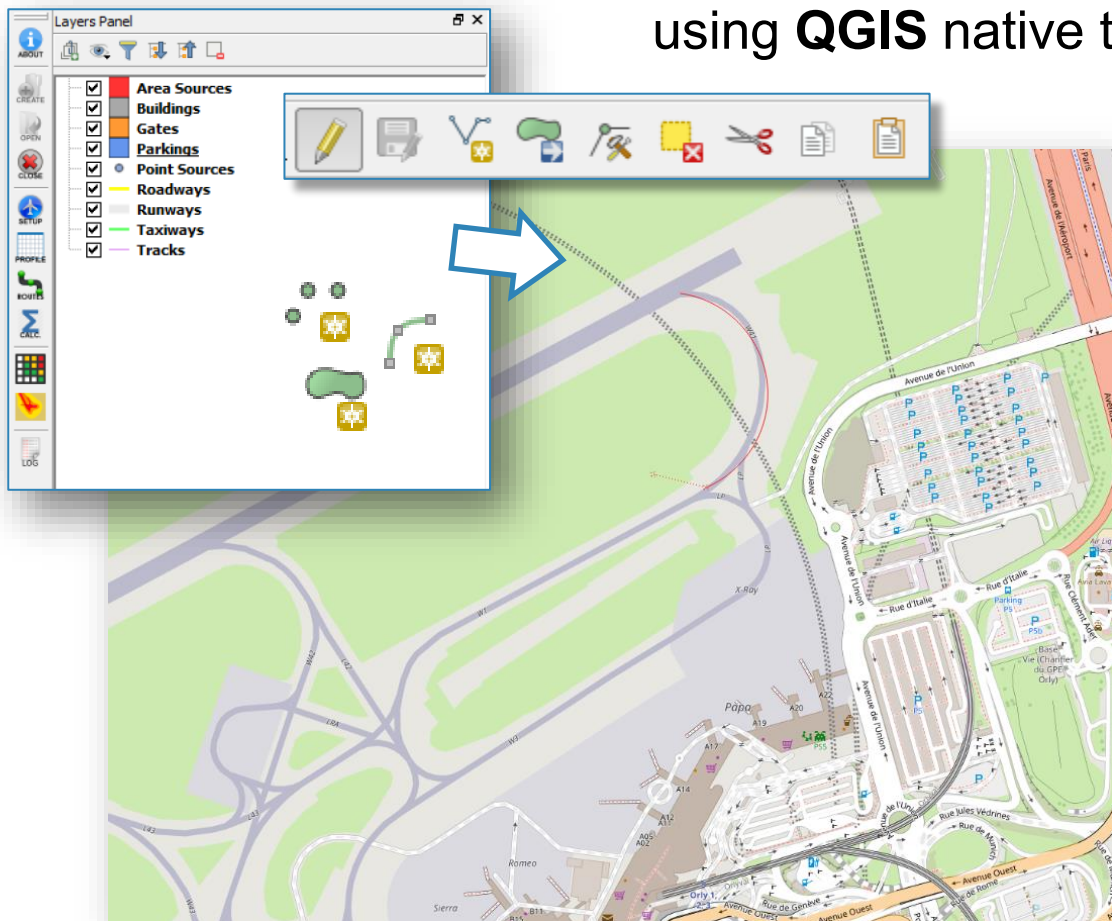


using **QGIS** native toolbar

		Tool	Purpose
	Turn on or off edit status of selected layer(s) based on the active layer status		
	Digitize using curve lines		
	Digitize polygon of regular shape		
	Add Feature: Capture Point		
	Add Feature: Capture Polygon		
	Vertex Tool (Current Layer)		
	Set whether the vertex editor panel should auto-open		
	Delete Selected features from the active layer		
	Copy selected Features from the active layer		
	Undo changes in the active layer		
	Redo changes in active layer		

Definition of Emission Sources

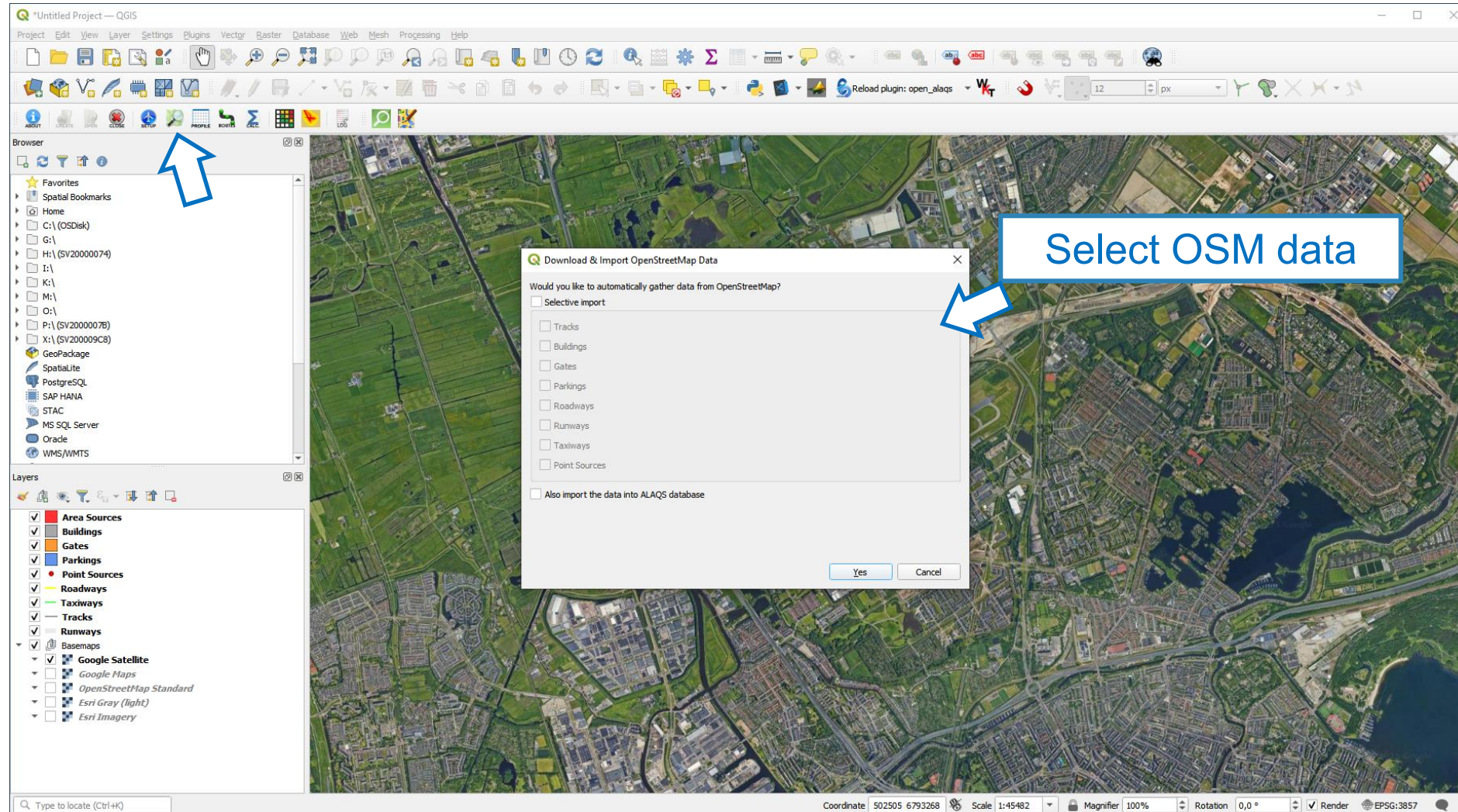
using **QGIS** native toolbar



Import OpenStreetMap data



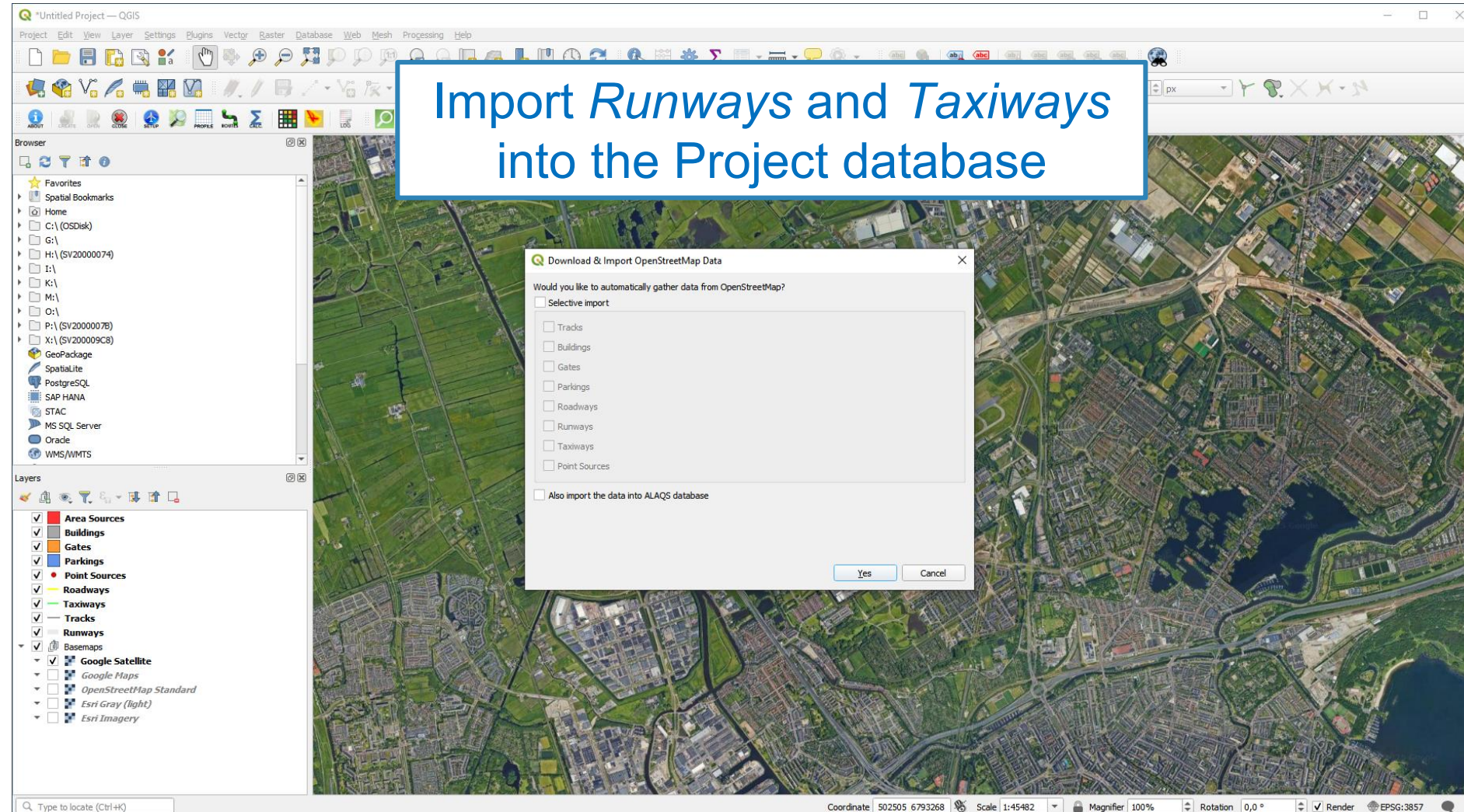
(Optional)



Import OpenStreetMap data



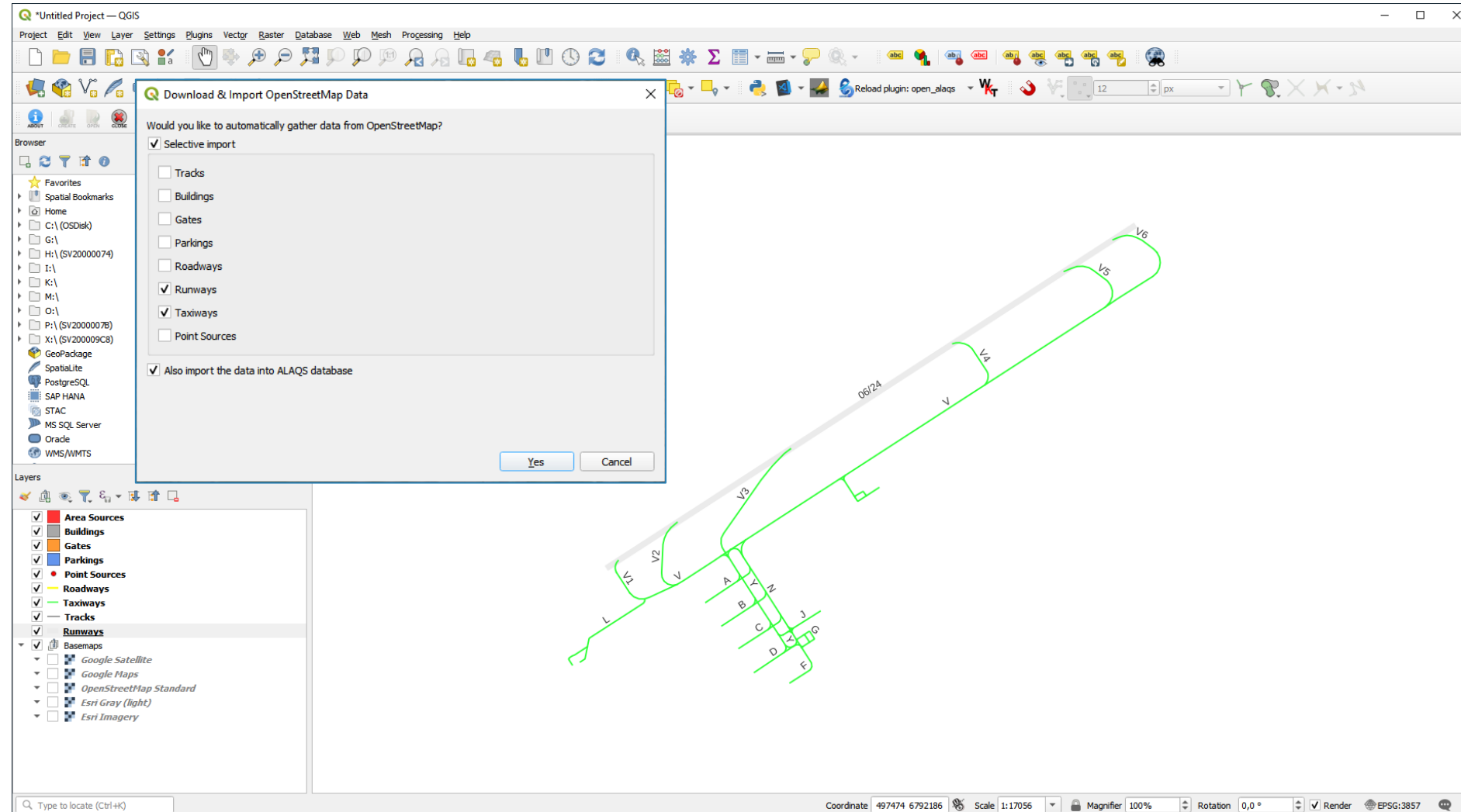
(Exercise)



Import OpenStreetMap data



(Exercise)



Import OpenStreetMap data



Check Attributes Table

Browser

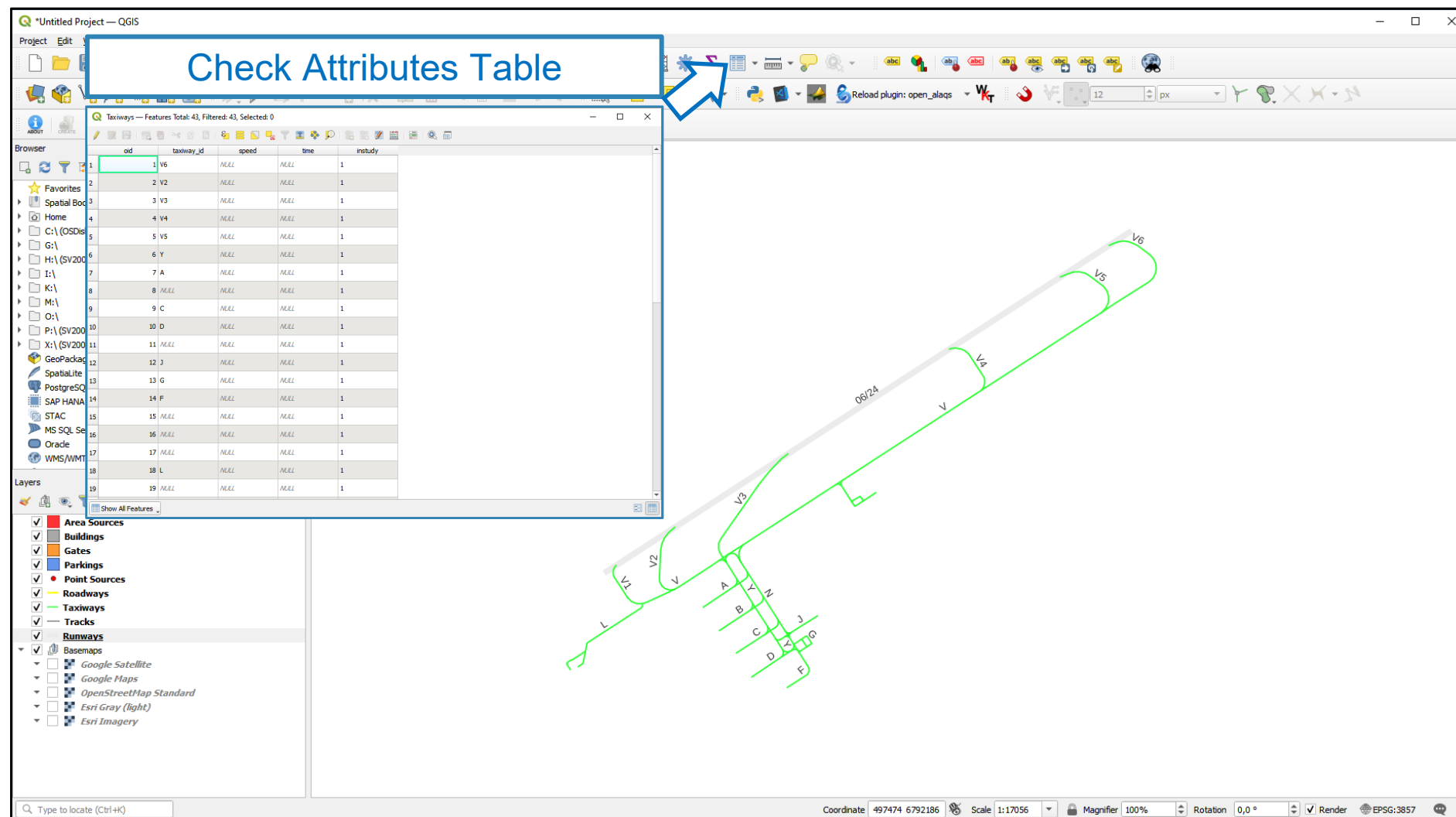
Taxiways — Features Total: 43, Filtered: 43, Selected: 0

oid	taxiway_id	speed	time	instudy
1	V6	NUL	NUL	1
2	V2	NUL	NUL	1
3	V3	NUL	NUL	1
4	V4	NUL	NUL	1
5	V5	NUL	NUL	1
6	Y	NUL	NUL	1
7	A	NUL	NUL	1
8	NUL	NUL	NUL	1
9	C	NUL	NUL	1
10	D	NUL	NUL	1
11	NUL	NUL	NUL	1
12	J	NUL	NUL	1
13	G	NUL	NUL	1
14	F	NUL	NUL	1
15	NUL	NUL	NUL	1
16	NUL	NUL	NUL	1
17	NUL	NUL	NUL	1
18	L	NUL	NUL	1
19	NUL	NUL	NUL	1

Layers

- ☒ Area Sources
- ☒ Buildings
- ☒ Gates
- ☒ Parkings
- ☒ Point Sources
- ☒ Roadways
- ☒ Taxiways
- ☒ Tracks
- ☒ Runways
- ☒ Basemaps
 - ☐ Google Satellite
 - ☐ Google Maps
 - ☐ OpenStreetMap Standard
 - ☐ Esri Gray (light)
 - ☐ Esri Imagery

Coordinate: 497474 6792186 Scale: 1:17056 Magnifier: 100% Rotation: 0,0° Render: EPSG:3857



Import OpenStreetMap data



Runways and
Taxiways

Ensure all imported Features have
valid attribute values and names

Taxiways — Features Total: 43, Filtered: 43, Selected: 0

	oid	taxiway_id	speed	time	instudy
1	1	V6	NULL	NULL	1
2	2	V2	NULL	NULL	1
3	3	V3	NULL	NULL	1
4	4	V4	NULL	NULL	1
5	5	V5	NULL	NULL	1
6	6	Y	NULL	NULL	1
7	7	A	NULL	NULL	1
8	8	NULL	NULL	NULL	1
9	9	C	NULL	NULL	1
10	10	D	NULL	NULL	1
11	11	NULL	NULL	NULL	1
12	12	J	NULL	NULL	1
13	13	G	NULL	NULL	1
14	14	F	NULL	NULL	1
15	15	NULL	NULL	NULL	1
16	16	NULL	NULL	NULL	1
17	17	NULL	NULL	NULL	1
18	18	L	NULL	NULL	1
19	19	NULL	NULL	NULL	1

Show All Features

Import OpenStreetMap data



Ensure all imported Features have valid attribute values and names

Taxiways — Features Total: 43, Filtered: 43, Selected: 0

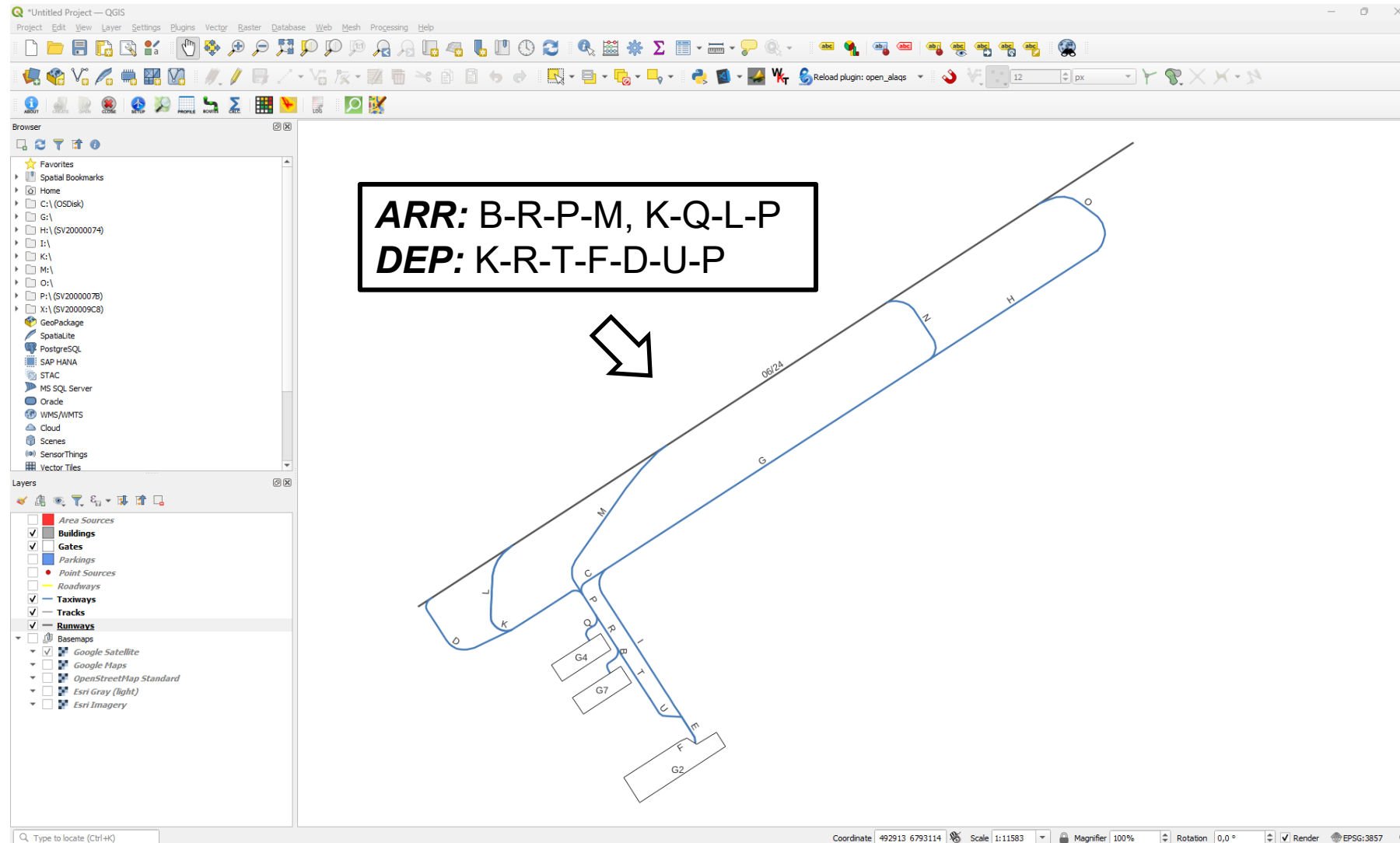
	oid	taxiway_id	speed	time	instudy
1	1	V6	NULL	NULL	1
2	2	V2	NULL	NULL	1
3	3	V3	NULL	NULL	1
4	4	V4	NULL	NULL	1
5	5	V5	NULL	NULL	1
6	6	Y	NULL	NULL	1
7	7	A	NULL	NULL	1
8	8	NULL	NULL	NULL	1
9	9	C	NULL	NULL	1
10	10	D	NULL	NULL	1
11	11	NULL	NULL	NULL	1
12	12	J	NULL	NULL	1
13	13	G	NULL	NULL	1
14	14	F	NULL	NULL	1
15	15	NULL	NULL	NULL	1
16	16	NULL	NULL	NULL	1
17	17	NULL	NULL	NULL	1
18	18	L	NULL	NULL	1
19	19	NULL	NULL	NULL	1

Show All Features

- Ensure that there are no duplicate IDs
- Typical speed: ~30 km/h
- Time: *calculated from speed and taxiway segment length*

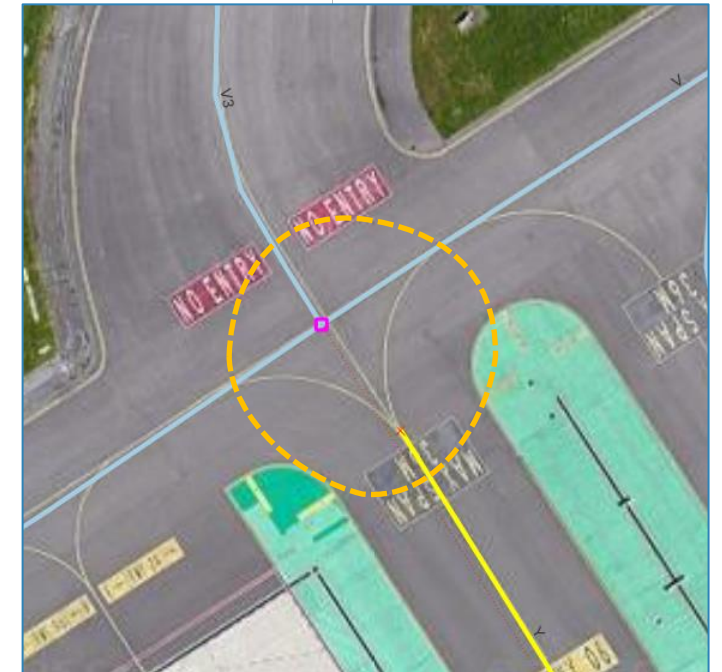
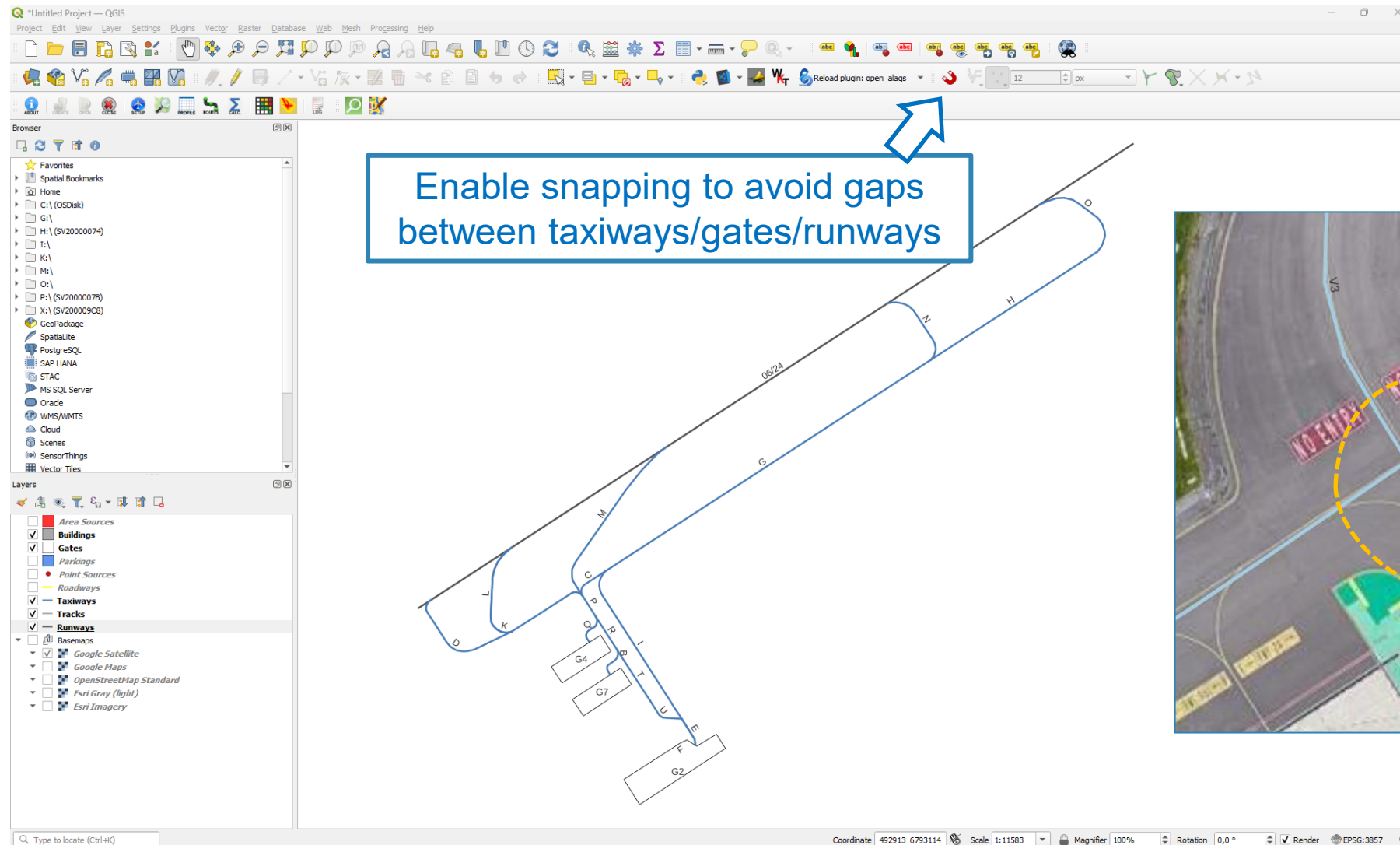


Create taxiways





Create taxiways

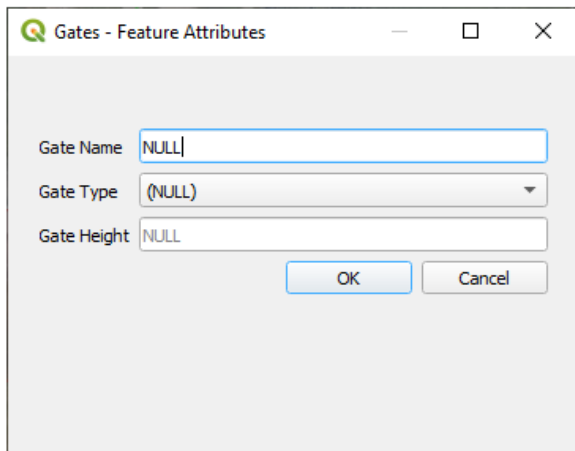


Add Gates

An airport gate refers to a designated location at an airport where aircraft park for boarding and disembarking passengers, loading/unloading cargo, and receiving services like refuelling, catering, and maintenance.

In OpenALAQS, gates are represented as polygons. Each gate can encompass several aircraft stands. The more stands grouped together within a single gate area, the less data preparation is needed (e.g., fewer taxi routes to define). However, if the gate area is too large, it might no longer accurately represent the location of the emissions.

Calculating gate emissions requires establishing the sum of four emission sources: GSE (Ground Support Equipment), GPU (Ground Power Unit), APU (Auxiliary Power Unit) and MES (Main Engine Start).



When adding a gate, the following information is required:

- Gate type (PIER, REMOTE or CARGO)
- Gate height not yet fully implemented

In OpenALAQS, GSE and GPU emissions factors, expressed in terms of grams of pollutant per hour, is assigned to each gate as a function of:

- The gate type (PIER, REMOTE or CARGO)
- The aircraft category (JET BUSINESS/REGIONAL/SMALL/MEDIUM/LARGE,TURBOPROPS,PISTON)
- The operation type (Arrival or Departure)

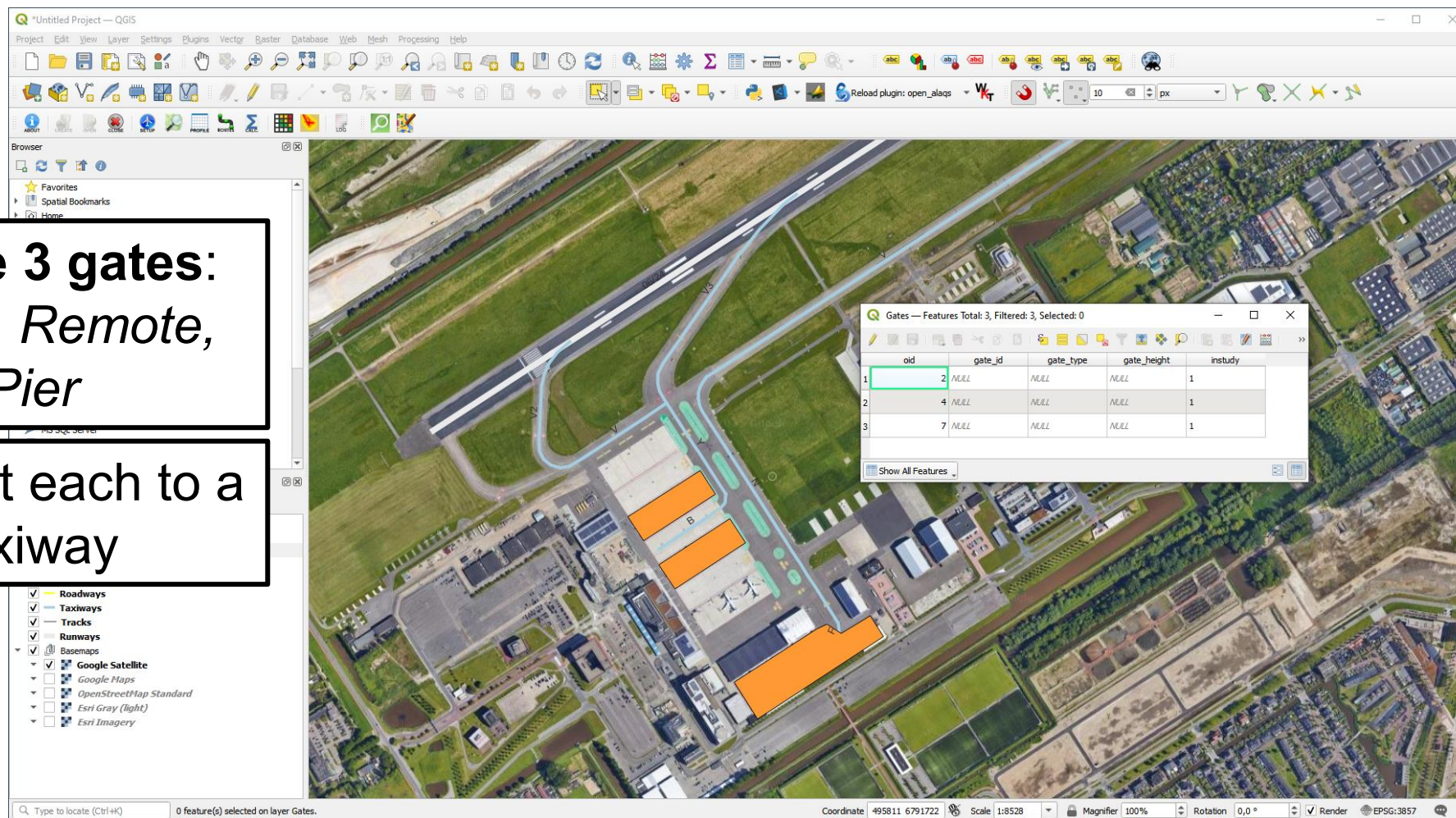
More information on the corresponding GSE/GPU, APU and MES emission factors and activity time is available in [OpenALAQS database](#).

Add Gates

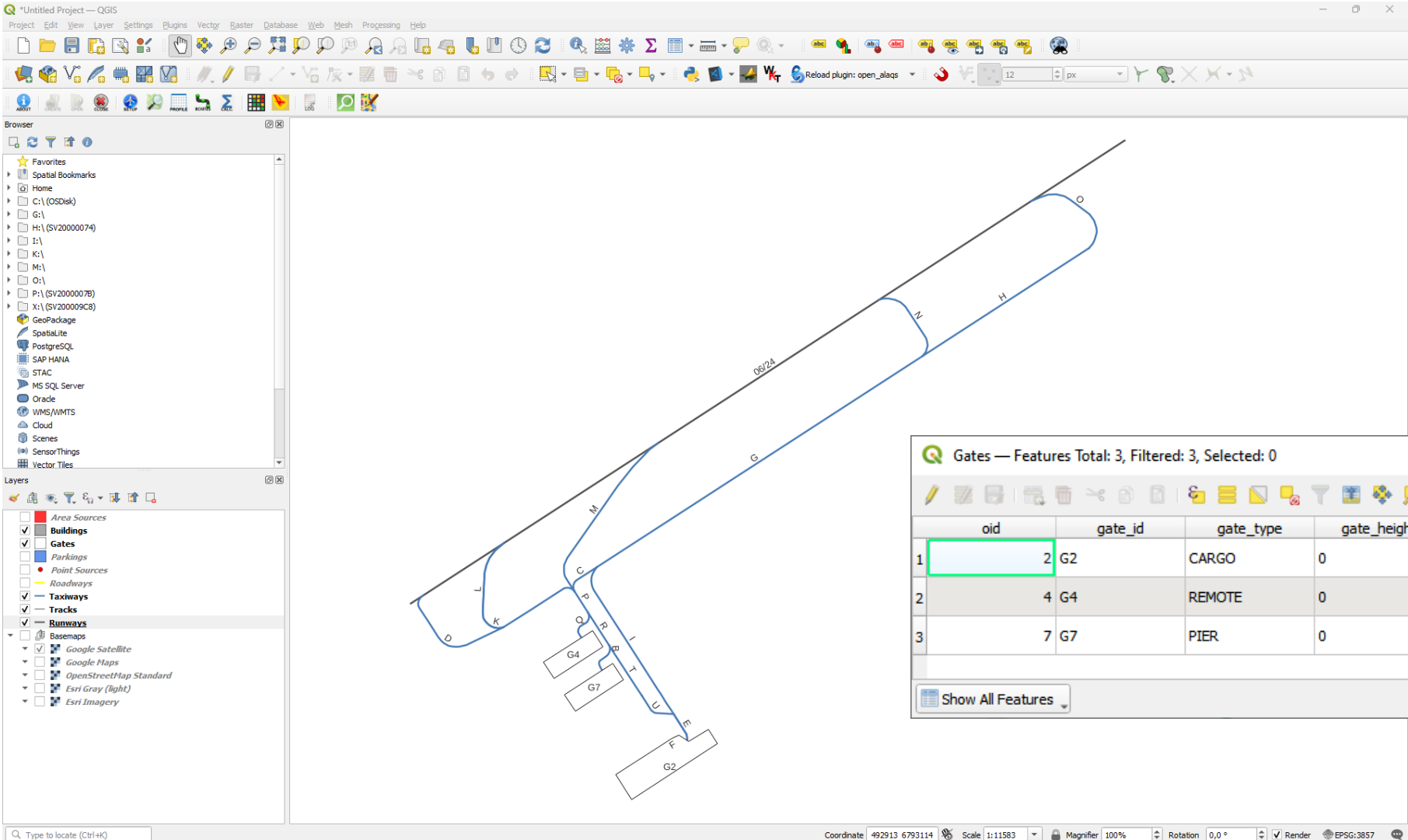


Create 3 gates:
*Cargo, Remote,
Pier*

**Connect each to a
taxiway**



Add Gates



The screenshot displays the QGIS desktop environment. The main map area shows an airport layout with various features labeled with letters (A through S). Three gates are highlighted with blue outlines and labeled G2, G4, and G7. The left sidebar contains a 'Browser' panel with a file tree and a 'Layers' panel with a list of map layers. The 'Layers' panel shows 'Buildings', 'Gates', 'Parkings', 'Point Sources', 'Roadways', 'Taxiways', 'Tracks', and 'Runways' as checked layers. The 'Gates' layer is selected. The bottom status bar shows the coordinate system as EPSG:3857, the scale as 1:11583, and the magnifier as 100%.

Gates — Features Total: 3, Filtered: 3, Selected: 0

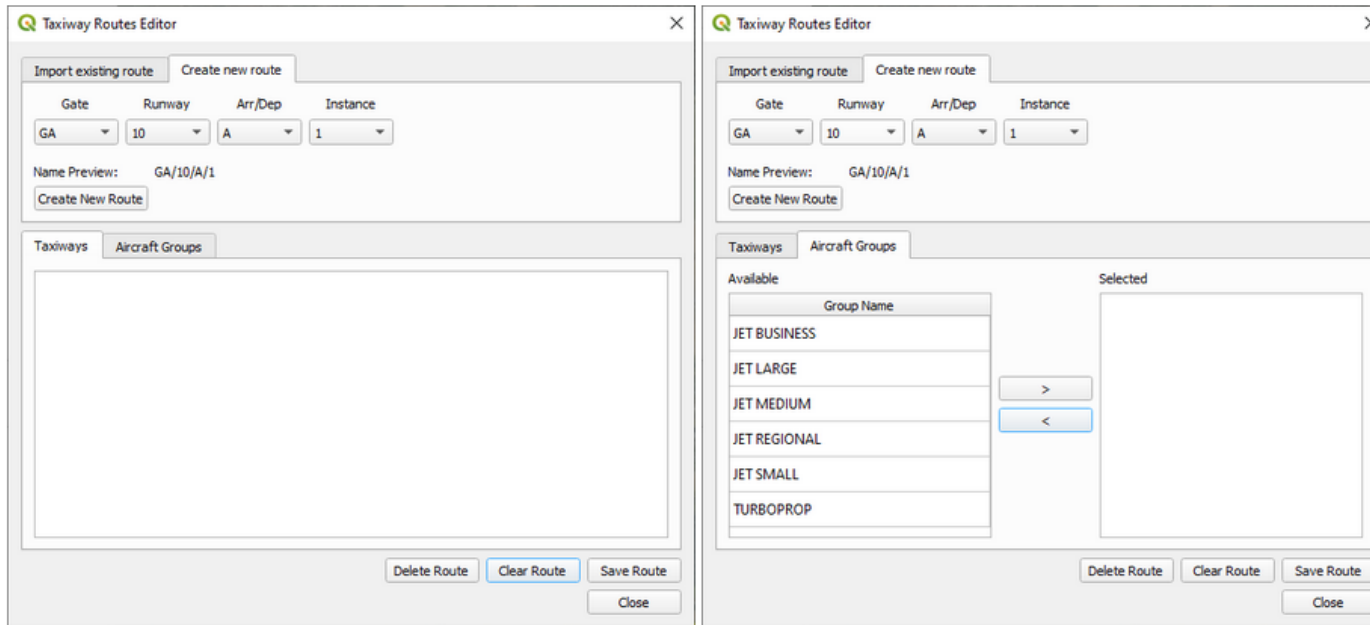
	oid	gate_id	gate_type	gate_height	instudy
1	2	G2	CARGO	0	1
2	4	G4	REMOTE	0	1
3	7	G7	PIER	0	1

Show All Features

Define Taxiway Routes

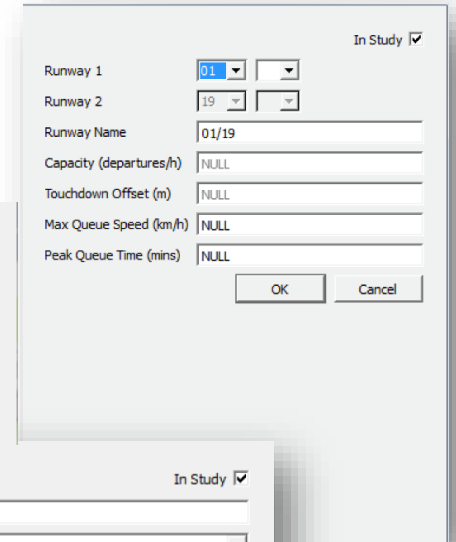
As explained in the section [Taxiways](#), taxi-routes describe the operational path of an aircraft between the runway and the gate (or vice versa).

Taxi-routes can be defined using the Taxiway Routes Editor.

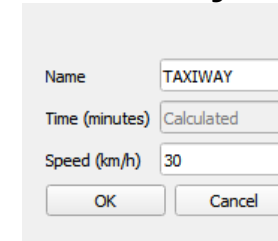


To define a taxi route in OpenALAQS, the user has to first create the taxi-route by selecting the gate, runway and operation type (arrival or departure). More than one taxi-routes can be defined for the same combination of gate, runway and operation type, using a different instance number. Once defined, the corresponding taxiway segments have to be selected together with the aircraft groups that can make use of the specific taxi-route.

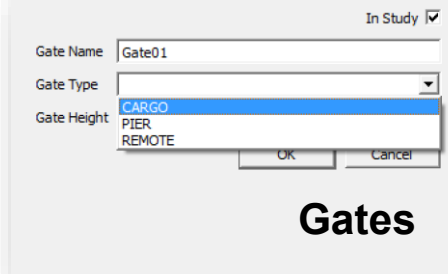
Runways



Taxiways



Gates



Define Taxiway Routes



Create 3 taxi routes: 2 *ARR*, 1 *DEP*

1 **Create New Route**

Select Gate, Runway,
Operation Type

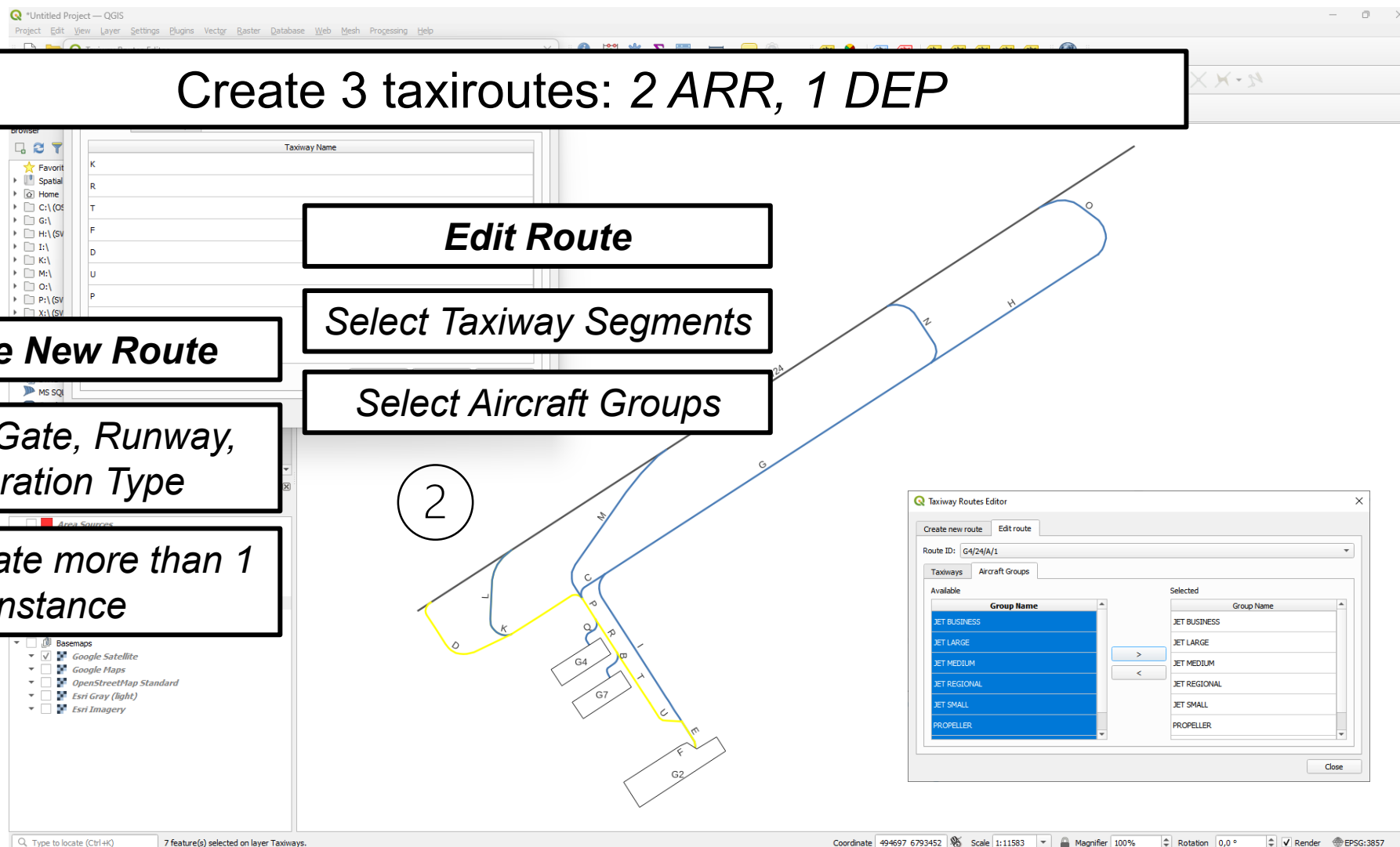
Can create more than 1
instance

Edit Route

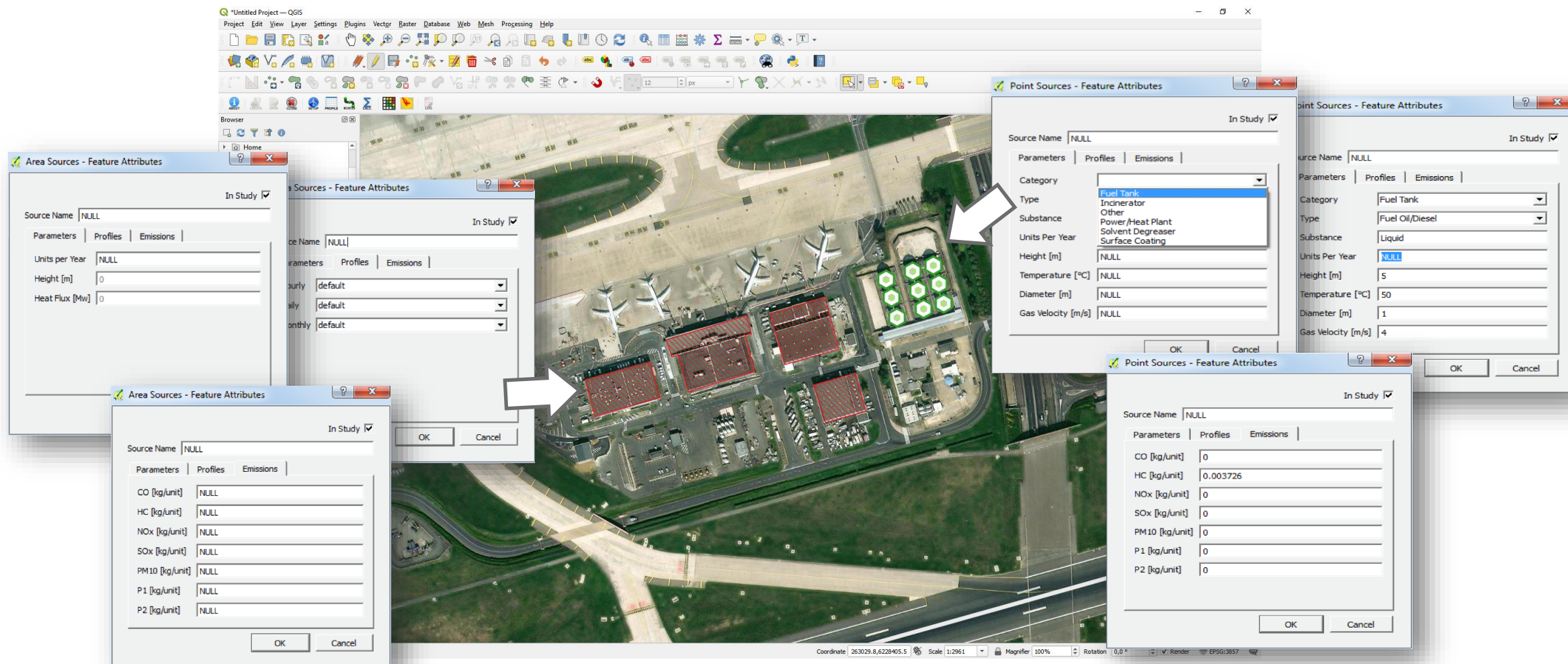
Select Taxiway Segments

Select Aircraft Groups

2



Stationary Sources



Custom (user-defined) EFs for each source

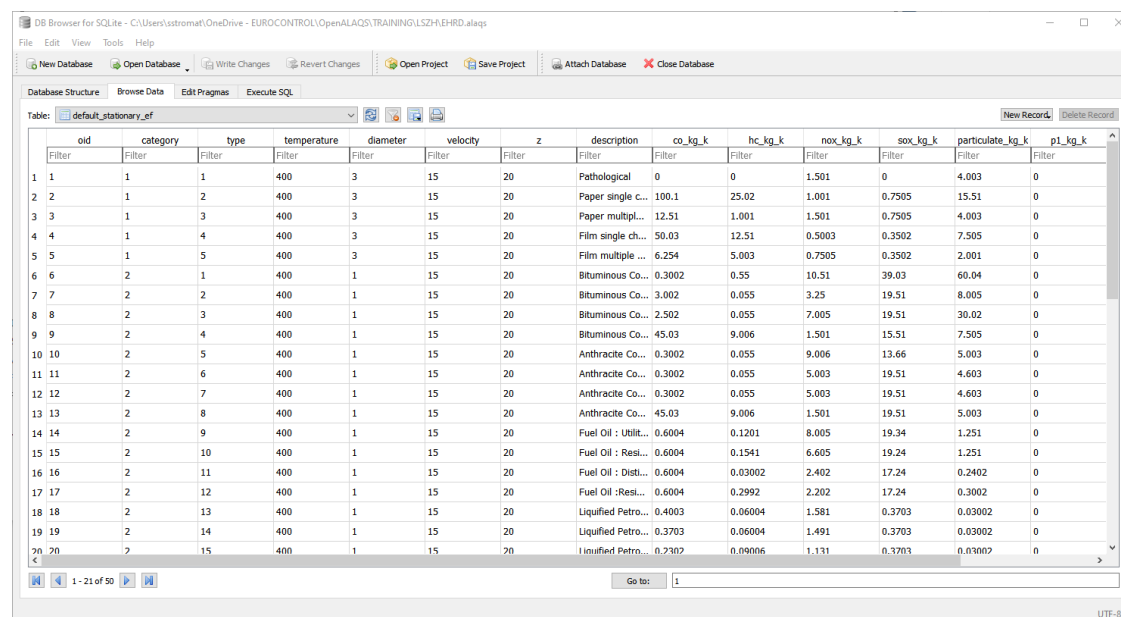
Default or custom EFs for each source

Stationary Sources



Create 1 point source and 1 area source

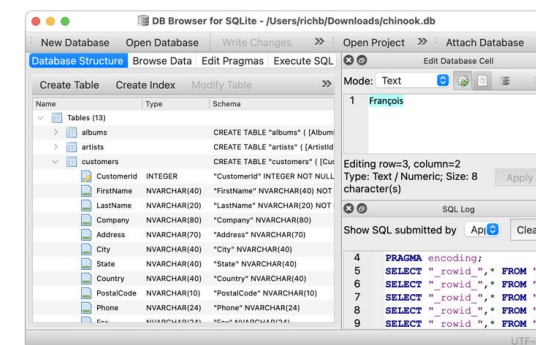
Modify the default emission factors for the point source category and type



oid	category	type	temperature	diameter	velocity	z	description	co_kg_k	hc_kg_k	nox_kg_k	sox_kg_k	particulate_kg_k	p1_kg_k
1	1	1	400	3	15	20	Pathological	0	0	1.501	0	4.003	0
2	1	2	400	3	15	20	Paper single c...	100.1	25.02	1.001	0.7505	15.51	0
3	1	3	400	3	15	20	Paper multipl...	12.51	1.001	1.501	0.7505	4.003	0
4	1	4	400	3	15	20	Film single ch...	50.03	12.51	0.5003	0.3502	7.505	0
5	1	5	400	3	15	20	Film multiple ...	6.254	5.003	0.7505	0.3502	2.001	0
6	2	1	400	1	15	20	Bituminous Co...	0.3002	0.55	10.51	39.03	60.04	0
7	2	2	400	1	15	20	Bituminous Co...	3.002	0.055	3.25	19.51	8.005	0
8	2	3	400	1	15	20	Bituminous Co...	2.502	0.055	7.005	19.51	30.02	0
9	2	4	400	1	15	20	Bituminous Co...	45.03	9.006	1.501	15.51	7.505	0
10	2	5	400	1	15	20	Anthracite Co...	0.3002	0.055	9.006	13.66	5.003	0
11	2	6	400	1	15	20	Anthracite Co...	0.3002	0.055	5.003	19.51	4.603	0
12	2	7	400	1	15	20	Anthracite Co...	0.3002	0.055	5.003	19.51	4.603	0
13	2	8	400	1	15	20	Anthracite Co...	45.03	9.006	1.501	19.51	5.003	0
14	2	9	400	1	15	20	Fuel Oil : Utilit...	0.6004	0.1201	8.005	19.34	1.251	0
15	2	10	400	1	15	20	Fuel Oil : Resi...	0.6004	0.1541	6.605	19.24	1.251	0
16	2	11	400	1	15	20	Fuel Oil : Dist...	0.6004	0.03002	2.402	17.24	0.2402	0
17	2	12	400	1	15	20	Fuel Oil : Resi...	0.6004	0.2992	2.202	17.24	0.3002	0
18	2	13	400	1	15	20	Liquidified Petro...	0.4003	0.06004	1.581	0.3703	0.03002	0
19	2	14	400	1	15	20	Liquidified Petro...	0.3703	0.06004	1.491	0.3703	0.03002	0
20	2	15	400	1	15	20	Liquidified Petro...	0.3702	0.09006	1.131	0.3703	0.03002	0

DB Browser for SQLite

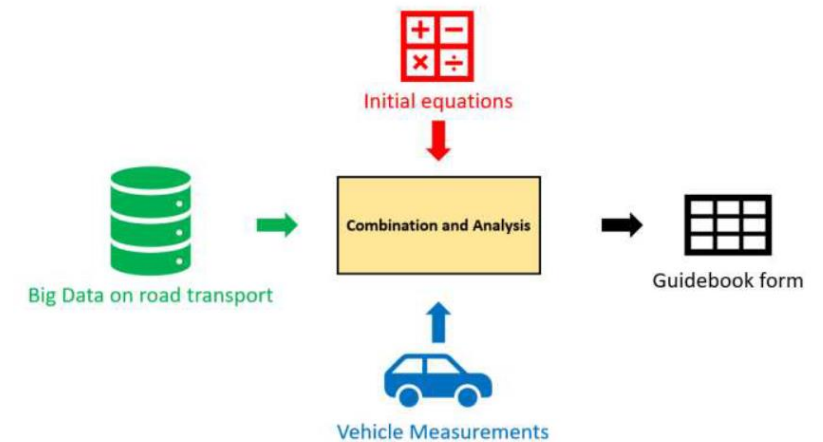
DB Browser for SQLite (DB4S) is a high quality, visual, open source tool designed for people who want to create, search, and edit SQLite or SQLitecipher database files. DB4S gives a familiar spreadsheet-like interface on the database in addition to providing a full SQL query facility. It works with Windows, macOS, and most versions of Linux and Unix. Documentation for the program is on the [wiki](#).



OpenALAQS files can be easily edited with DB Browser for SQLite an open-source tool designed for manipulating SQLite database files.

Roadway traffic

- ❑ Estimation of **Roadway traffic emissions** (landside, airside, parking lots) is based on **COPERT** (version V)
- ❑ COPERT contains **emission factors for more than 450 individual vehicle types** (e.g. PC, LDV, HDV)
- ❑ Estimates **emissions from all relevant road vehicle operation modes**:
 - ❑ 'hot' & 'cold start' emissions
 - ❑ non-exhaust emissions (from fuel evaporation, tyre and brake wear emissions)
- ❑ The development of COPERT is **coordinated by the European Environment Agency (EEA)**



<https://www.emisia.com>

Roadway traffic

Parking Lots

Emissions from parking areas for vehicles are estimated based on the [COPERT](#) methodology.

When adding an airport parking lot, the following information is required:

- **Parameters**
 - **Number per year:** Total number of vehicles per year
 - **Height:** Height at which emissions are released (in meters) not yet fully implemented
 - **Speed:** Average travel speed in parking (in km/h)
 - **Travel distance:** Average travel distance in parking (in meters)
 - **Idle time:** Vehicle average idling time between entry and exit (in minutes)
 - **Average parking time:** Average time a vehicle remains on parking (in minutes)
- **Profiles**
 - Hourly, Daily or Monthly activity profiles
- **Fleet mix**
 - **PC (Petrol) [in %]:** Passenger Cars (Petrol)
 - **PC (Diesel) [in %]:** Passenger Cars (Diesel)
 - **LDV (Petrol) [in %]:** Light Duty Vehicles (Petrol)
 - **LDV (Diesel) [in %]:** Light Duty Vehicles (Diesel)
 - **HDV (Petrol) [in %]:** Heavy Duty Vehicles (Petrol)
 - **HDV (Diesel) [in %]:** Heavy Duty Vehicles (Diesel)
 - **Motorcycles [in %]**
 - **Buses [in %]**

Roadways

Airside or landside emissions are calculated using the same methodology as described above.

When adding a roadway, the following information is required:

- **Parameters**
 - **Movements per year:** Number of annual movements
 - **Height:** Height at which emissions are released (in meters) not yet fully implemented
 - **Speed:** Vehicles speed in roadway (in km/h)
- **Profiles:**
 - Hourly, Daily or Monthly activity profiles
- **Fleet mix**
 - **PC (Petrol) [in %]:** Passenger Cars (Petrol)
 - **PC (Diesel) [in %]:** Passenger Cars (Diesel)
 - **LDV (Petrol) [in %]:** Light Duty Vehicles (Petrol)
 - **LDV (Diesel) [in %]:** Light Duty Vehicles (Diesel)
 - **HDV (Petrol) [in %]:** Heavy Duty Vehicles (Petrol)
 - **HDV (Diesel) [in %]:** Heavy Duty Vehicles (Diesel)
 - **Motorcycles [in %]**
 - **Buses [in %]**

Roadway traffic



Create 1 parking lot and 1 roadway

1

Create Parking Lot

*Provide values for all
necessary parameters*

Calculate EFs

2

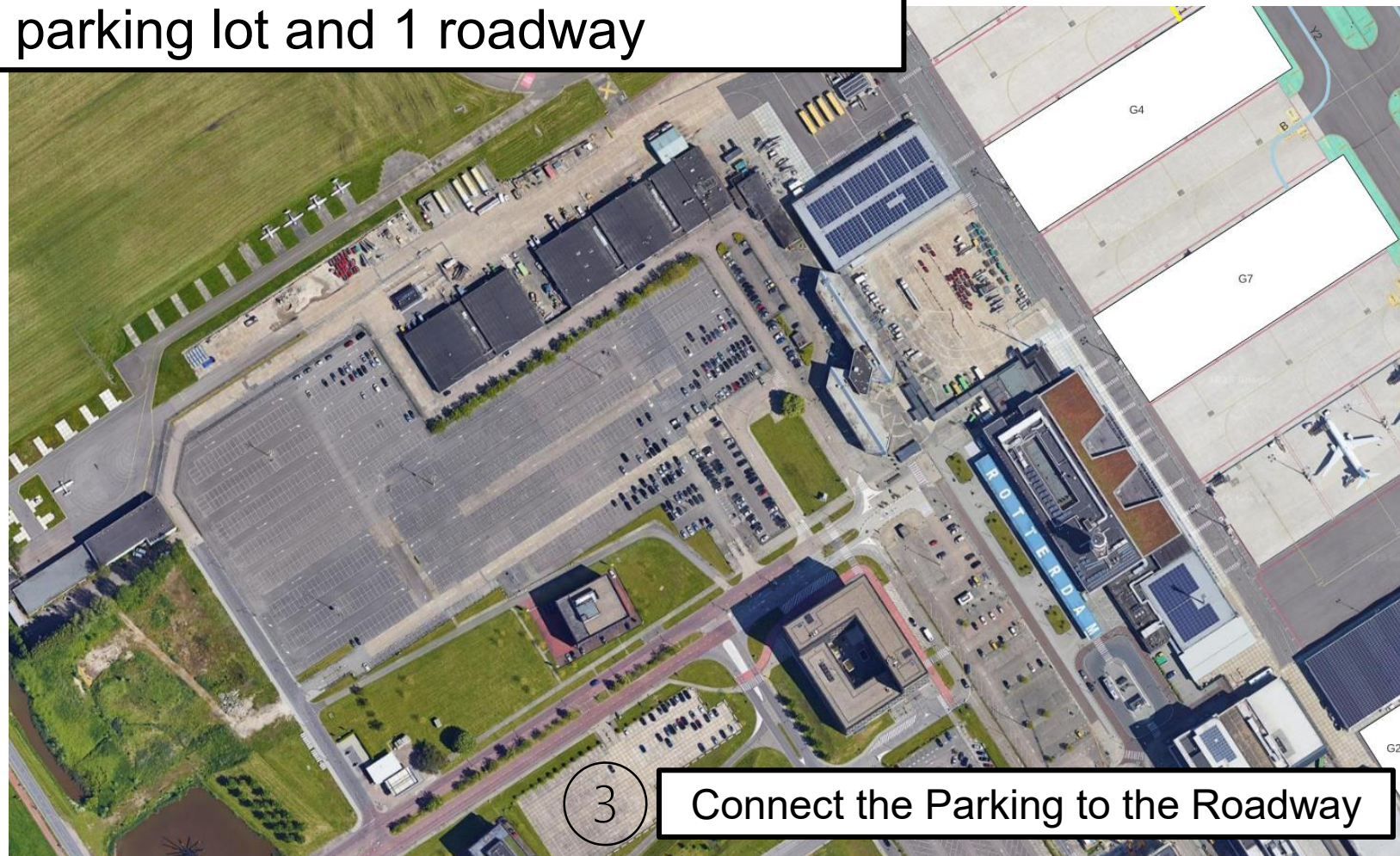
Create Roadway

*Provide values for all
necessary parameters*

Calculate EFs

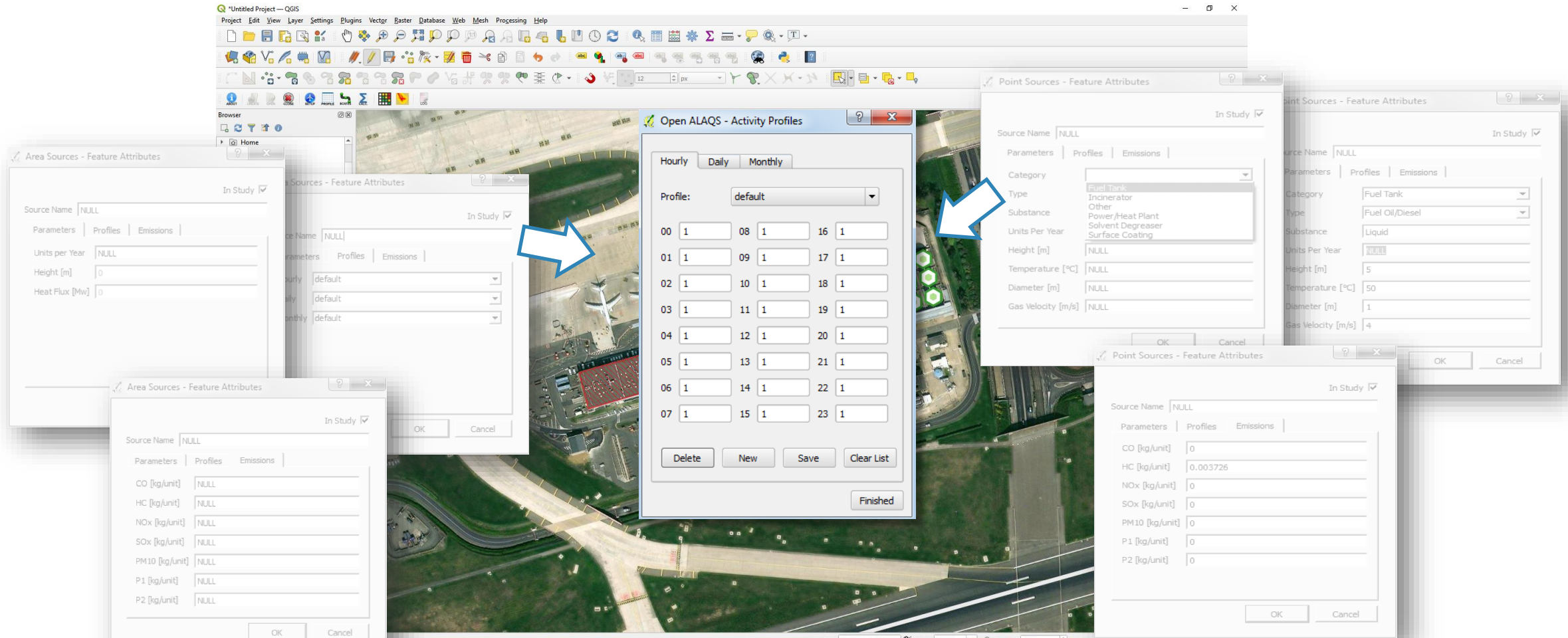
3

Connect the Parking to the Roadway



Activity Profiles

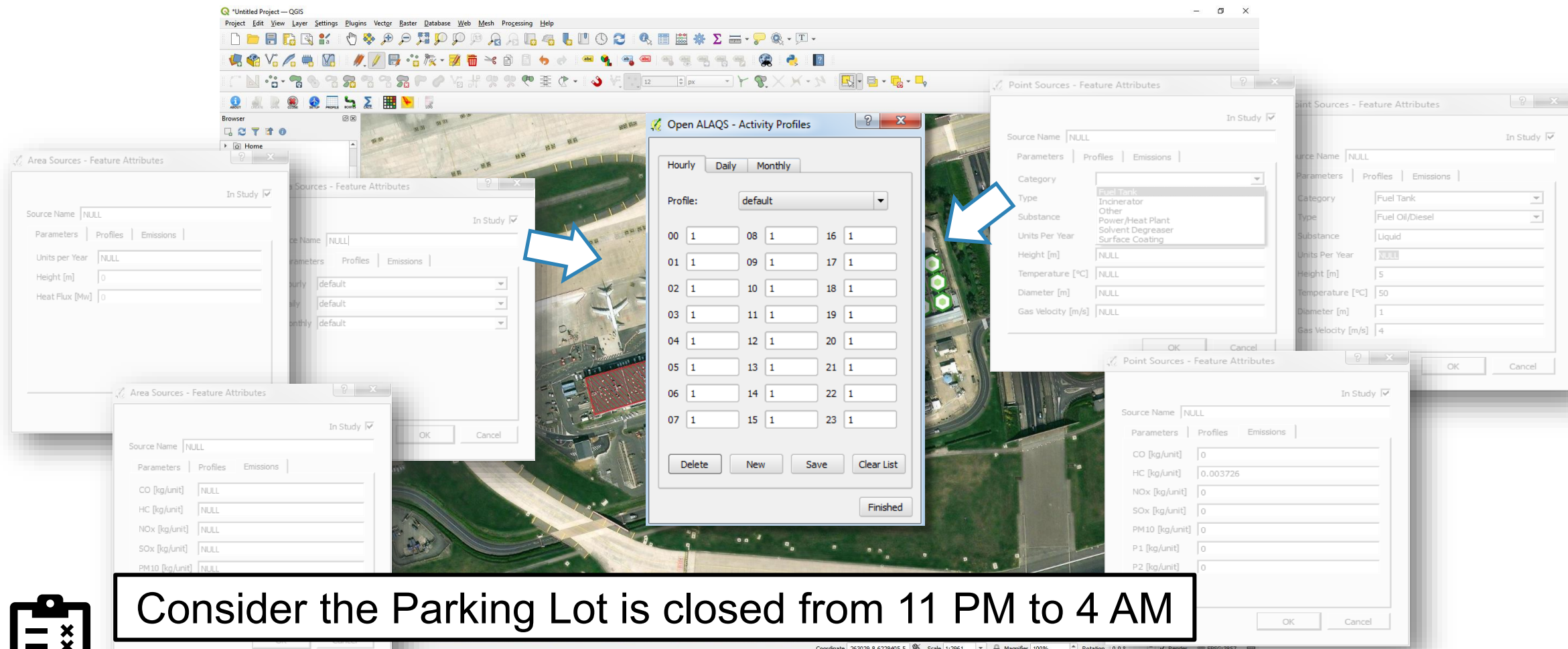
(Optional)



Each activity multiplier is a decimal number, between 0 and 1.
The default profile values are 1 (i.e., 100%) meaning the emission source is fully active.

Activity Profiles

(Optional)



Area Sources - Feature Attributes

Source Name: NULL

Parameters | Profiles | Emissions

Units Per Year: NULL

Height [m]: 0

Heat Flux [Mw]: 0

Open ALAQS - Activity Profiles

Hourly | Daily | Monthly

Profile: default

00	1	08	1	16	1
01	1	09	1	17	1
02	1	10	1	18	1
03	1	11	1	19	1
04	1	12	1	20	1
05	1	13	1	21	1
06	1	14	1	22	1
07	1	15	1	23	1

Delete New Save Clear List Finished

Point Sources - Feature Attributes

Source Name: NULL

Parameters | Profiles | Emissions

Category: Fuel Tank

Type: Fuel Oil/Diesel

Substance: Liquid

Units Per Year: 1000

Height [m]: 5

Temperature [°C]: 50

Diameter [m]: 1

Gas Velocity [m/s]: 4

Area Sources - Feature Attributes

Source Name: NULL

Parameters | Profiles | Emissions

CO [kg/unit]: NULL

HC [kg/unit]: NULL

NOx [kg/unit]: NULL

SOx [kg/unit]: NULL

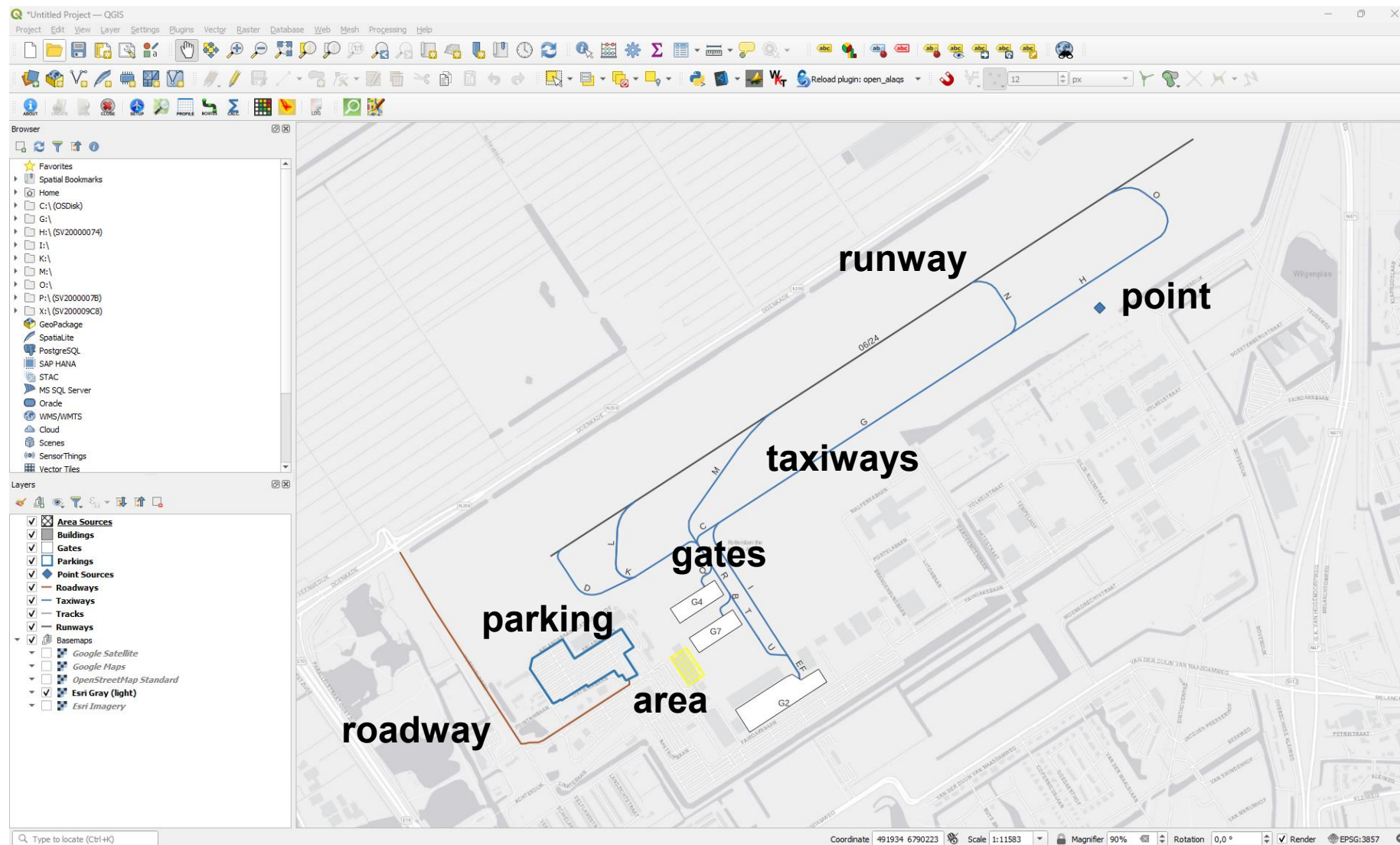
PM10 [kg/unit]: NULL



Consider the Parking Lot is closed from 11 PM to 4 AM

Create an Activity Profile and apply it to the Parking Lot

Emission Sources



Generate Emissions Inventory – Input Data

Generate Emission Inventory

Aircraft movements

1

Emission Inventory Output

Directory:

File Name:

Movement Data

Movements Table:

Filter Start Date:

Filter End Date:

Meteorological Data

Meteorological Table:

Model domain & resolution

Modelled Domain

X Resolution:

x

40 cells

Resolution:

x

40 cells

Z Resolution:

x

40 cells

Advanced Options

Method:

Towing Speed:

Vertical Limit:

Scenario	DateTime(YYYY-mm-dd hh:mm:ss)	Temperature(K)	Humidity(kg_water/kg_dry_air)	RelativeHumidity(%)	SeaLevelPressure(mb)	WindSpeed(m/s)	WindDirection(degrees)	ObukhovLength(m)	MixingHeight(m)
default	2004-01-01 00:00:00	276.48	0.00436	0.93	100960	0.89	280	-99999	914.4
default	2004-01-01 06:00:00	275.93	0.00406	0.93	100960	0.89	140	6.2	914.4
default	2004-01-01 12:00:00	279.82	0.0047	0.81	100540	6.26	180	-8888	914.4
default	2004-01-01 18:00:00	279.82	0.00588	1	99270	9.83	180	1954.1	914.4
default	2004-01-02 00:00:00	281.48	0.00551	0.81	98740	6.26	220	436.9	914.4
default	2004-01-02 06:00:00	282.59	0.00637	0.87	98110	5.81	180	647.4	914.4
default	2004-01-02 12:00:00	280.93	0.00519	0.81	97710	4.02	240	-5268.3	914.4
default	2004-01-02 18:00:00	281.48	0.00476	0.71	99280	4.02	250	-99999	914.4

2

runway_time	block_time	aircraft_registration	aircraft	gate	departure_arrival	runway	engine_name	prof_id	track_id	taxi_route	tow_ratio	apu_code	taxi_engine_count
2015-01-01 05:18:00	2015-01-01 05:23:00		B744	T01_T19	A	24	1PW042	747400-A-1			1		1
2015-01-01 06:27:00	2015-01-01 06:31:30		MD11	T01_T19	A	24	1PW059	MD11GE-A-1			1		1
2015-01-01 07:20:00	2015-01-01 07:25:00		MD11	T01_T19	A	24	1PW059	MD11GE-A-1			1		1
2015-01-01 08:17:00	2015-01-01 08:08:30		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1
2015-01-01 09:21:38	2015-01-01 09:25:21		T154	T01_T19	A	24	1KK002	727D17-A-1			1		1
2015-01-01 09:39:00	2015-01-01 09:27:00		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1
2015-01-01 10:30:16	2015-01-01 10:34:18		E145	T01_T19	A	24	6AL006	CL601-A-1			1		1
2015-01-01 10:44:08	2015-01-01 10:38:40		T154	T01_T19	D	24	1KK002	727D17-D-3			1		1
2015-01-01 11:02:29	2015-01-01 11:07:12		A319	T20_T29	A	24	3CM028	A319-A-1			1		1
2015-01-01 11:52:00	2015-01-01 11:43:30		MD11	T01_T19	D	24	1PW059	MD11GE-D-7			1		1

Meteorology

Status: Ready

Create Inventory

Close

Movements Table

This table contains all the aircraft movements occurring at the airport during a certain period. It must include the following information:

- **runway_time**: Date and time of arrival at the airport runway (YYYY-MM-DD HH:MM:SS)
- **block_time**: Date and time of arrival at the gate (YYYY-MM-DD HH:MM:SS)
- **aircraft**: The ICAO aircraft type
- **gate**: The stand used for the aircraft operation
- **departure_arrival**: The type of operation (arrival or departure)
- **runway**: The name of the runway used for the aircraft operation
- **taxi_route**: the name of the taxi route used for the aircraft operation
- **apu_code**: code defining APU usage away from gate/stand (0: APU used at gate only, 1: APU on while taxiing, 2: APU on during taxiing, take-off, climb or approach)

The rest of the parameters are not mandatory and can be left empty. They will only be used if the user provides specific values. Otherwise, the default values from the internal database will be used.

- **aircraft_registration**: aircraft registration number not yet fully implemented
- **engine_name**: engine name identifier (from the internal database)
- **profile_id**: performance profile identifier (from the internal database)
- **track_id**: aircraft trajectory identifier not yet fully implemented
- **tow_ratio**: take-off gross weight divided by maximum take-off weight (float, <=1) not yet fully implemented
- **taxi_engine_count**: number of engines used during taxing (integer)
- **set_time_of_main_engine_start_after_block_off_in_s**: time spent for MES after leaving the gate (in seconds)
- **set_time_of_main_engine_start_before_takeoff_in_s**: time spent for MES before take-off (in seconds)
- **set_time_of_main_engine_off_after_runway_exit_in_s**: time spent for MES after leaving the runway exit (in seconds)
- **engine_thrust_level_for_taxiing**: taxing thrust level setting (ICAO default: 7%) not yet fully implemented
- **taxi_fuel_ratio**: ratio between actual fuel flow and fuel for the idle mode (float) not yet fully implemented
- **number_of_stop_and_gos**: number of stop and go points during taxing not yet fully implemented
- **domestic**: domestic/international flight (Y or N) not yet fully implemented

(Optional)

Movements Table

This table contains all the aircraft movements occurring at the airport during a certain period. It must include the following information:

- **runway_time**: Date and time of arrival at the airport runway (YYYY-MM-DD HH:MM:SS)

Use the provided example file to create a Movements Table .csv file for your study

Create at least 1 ARR and 1 DEP for each gate and runway direction

Consider different AC types

- **profile_id**: performance profile identifier (from the internal database)
- **track_id**: aircraft trajectory identifier not yet fully implemented
- **tow_ratio**: take-off gross weight divided by maximum take-off weight (float, <=1) not yet fully implemented
- **taxi_engine_count**: number of engines used during taxing (integer)
- **set_time_of_main_engine_start_after_block_off_in_s**: time spent for MES after leaving the gate (in seconds)
- **set_time_of_main_engine_start_before_takeoff_in_s**: time spent for MES before take-off (in seconds)
- **set_time_of_main_engine_off_after_runway_exit_in_s**: time spent for MES after leaving the runway exit (in seconds)
- **engine_thrust_level_for_taxiing**: taxing thrust level setting (ICAO default: 7%) not yet fully implemented
- **taxi_fuel_ratio**: ratio between actual fuel flow and fuel for the idle mode (float) not yet fully implemented
- **number_of_stop_and_gos**: number of stop and go points during taxing not yet fully implemented
- **domestic**: domestic/international flight (Y or N) not yet fully implemented

(Optional)



Meteorological Data

OpenALAQS requires certain meteorological data in order to accurately calculate emissions using advanced methods (e.g. BFFM2 method, correction of NO_x for ambient conditions) but also for the dispersion calculation.

The required meteorological parameters are:

- **Scenario:** simulation name identifier
- **DateTime** (in YYYY-MM-DD HH:MM:SS)
- **Temperature** (in K)
- **Humidity** (in kg water/kg dry air)
- **RelativeHumidity** (in %)
- **SeaLevelPressure** (in Pa)

open
alaqs

- **WindSpeed** (in m/s)
- **WindDirection** (in degrees)
- **ObukhovLength** (in m)
- **MixingHeight** (in m)

AUSTAL

Input data may come from local or national data providers (e.g. METAR, SYNOP) or reanalysis data. The compilation of the meteorological data file for OpenALAQS/AUSTAL2000 is left to the user. Otherwise, default ISA conditions values are considered for all necessary meteorological parameters.

Generate Emissions Inventory – Input Data



Generate Emission Inventory

Emission Inventory Output

Directory:

File Name:

Movement Data

Movements Table:

Filter Start Date: 2000-01-01 00:00:00

Filter End Date: 2000-01-02 00:00:00

Meteorological Data

Meteorological Table:

Modelled Domain

X Resolution: 250 m

Y Resolution: 250 m

Z Resolution: 250 m

Advanced Options

Method: ALAQS

Towing Speed: 10,00 km/h

Vertical Limit: 914,40 m

Status: Ready

Create Inventory

Close

Select a directory and a file name for your output .alaqs file

Use the csv file you created for your study

Use the provided MET data

Define the size & resolution of your domain

runway_time	block_time	aircraft_registration	aircraft	gate	departure_arrival	runway	engine_name	prof_id	track_id	taxi_route	tow_ratio	apu_code	taxi_engine_count
								7400-A-1			1		1
								011GE-A-1			1		1
								011GE-A-1			1		1
								011GE-D-7			1		1
								7D17-A-1			1		1
								011GE-D-7			1		1
2015-01-01 10:30:16	2015-01-01 10:34:18		E145	T01 T19 A		24	6AL006	CL601-A-1			1		1
								7D17-D-3			1		1
								19-A-1			1		1
								011GE-D-7			1		1

								WindSpeed(m/s)	WindDirection(degrees)	ObukhovLength(m)	MixingHeight(m)
								0.89	280	-99999	914.4
								0.89	140	6.2	914.4
								6.26	180	-8888	914.4
								9.83	180	1954.1	914.4
								6.26	220	436.9	914.4
								5.81	180	647.4	914.4
								4.02	240	-5268.3	914.4
								4.02	250	-99999	914.4

Calculate and Visualise Emissions

☐ Calculate emissions:

- ☐ Per Source Type
- ☐ Per Source Name
- ☐ Per Pollutant

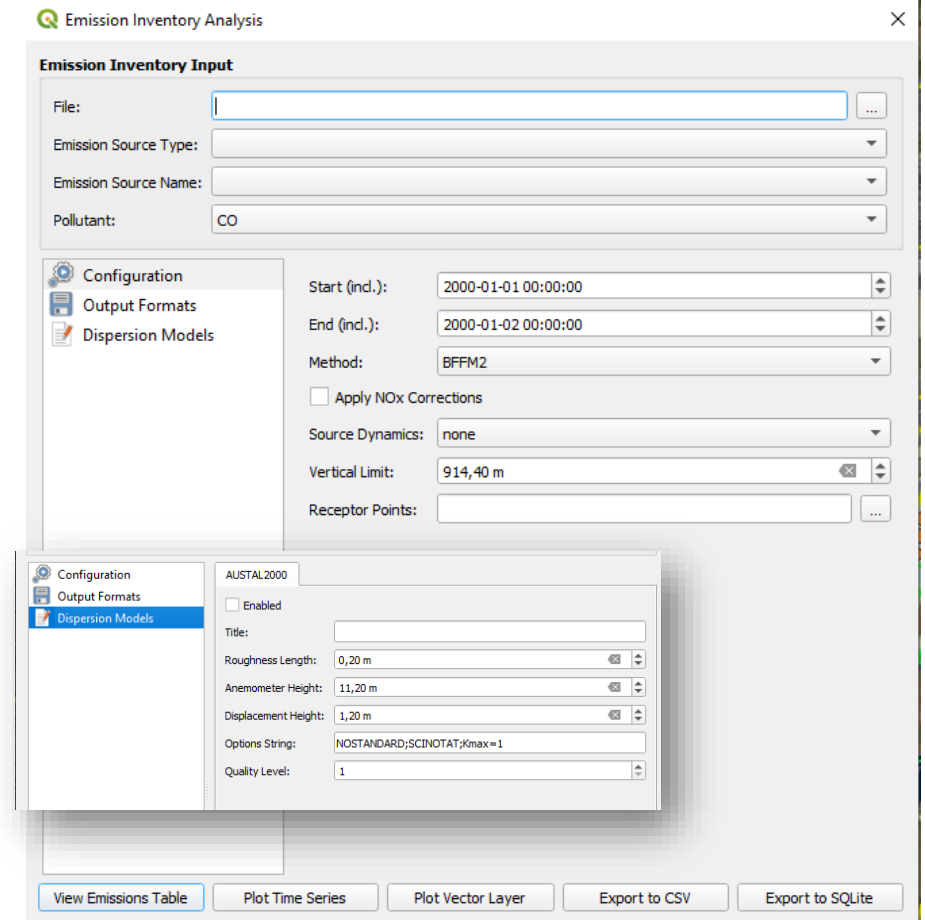


☐ Dispersion module can be activated/deactivated and configured from within Open-ALAQs



☐ Various ways of visualising/exporting the results

- ☐ Emissions are aggregated and presented for each hour of the simulation time period



The screenshot displays the 'Emission Inventory Analysis' software interface. The main window is titled 'Emission Inventory Input' and contains several input fields: 'File:', 'Emission Source Type:', 'Emission Source Name:', and 'Pollutant:' (set to 'CO'). Below these is a 'Configuration' section with a sidebar showing 'Configuration', 'Output Formats', and 'Dispersion Models'. The 'Dispersion Models' section is active, showing 'Start (ind.): 2000-01-01 00:00:00', 'End (ind.): 2000-01-02 00:00:00', 'Method: BFFM2', 'Apply NOx Corrections' (unchecked), 'Source Dynamics: none', 'Vertical Limit: 914,40 m', and 'Receptor Points:'. A smaller window titled 'AUSTAL2000' is open, showing 'Enabled' (unchecked), 'Title:', 'Roughness Length: 0,20 m', 'Anemometer Height: 11,20 m', 'Displacement Height: 1,20 m', 'Options String: NOSTANDARD;SCINOTAT;Kmax=1', and 'Quality Level: 1'. At the bottom of the main window are buttons for 'View Emissions Table', 'Plot Time Series', 'Plot Vector Layer', 'Export to CSV', and 'Export to SQLite'.

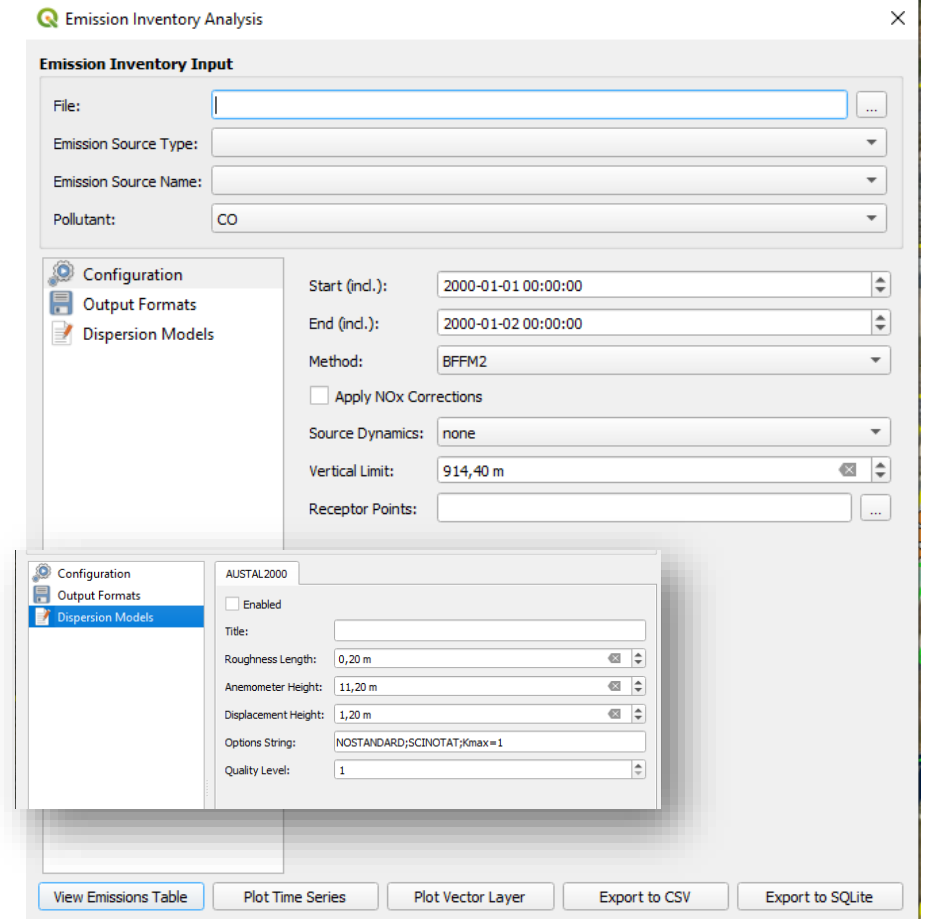
Calculate and Visualise Emissions



Select the output .alaqs file you've created before

Calculate the Parking Lot CO emissions

Visualise the calculated emissions as a Table and Vector Layer



The screenshot shows the 'Emission Inventory Analysis' software window. The main panel is titled 'Emission Inventory Input' and contains fields for 'File:', 'Emission Source Type:', 'Emission Source Name:', and 'Pollutant:' (set to 'CO'). Below this is a 'Configuration' section with a sidebar containing 'Configuration', 'Output Formats', and 'Dispersion Models'. The 'Dispersion Models' section is active, showing 'Start (incl.): 2000-01-01 00:00:00', 'End (incl.): 2000-01-02 00:00:00', 'Method: BFFM2', 'Apply NOx Corrections' (unchecked), 'Source Dynamics: none', 'Vertical Limit: 914,40 m', and 'Receptor Points:'. A smaller window titled 'AUSTAL2000' is open, showing 'Enabled' (unchecked), 'Title:', 'Roughness Length: 0,20 m', 'Anemometer Height: 11,20 m', 'Displacement Height: 1,20 m', 'Options String: NOSTANDARD;SCINOTAT;Kmax=1', and 'Quality Level: 1'. At the bottom of the main window are buttons for 'View Emissions Table', 'Plot Time Series', 'Plot Vector Layer', 'Export to CSV', and 'Export to SQLite'.

Aircraft emissions

❑ Different options for calculating aircraft emissions:

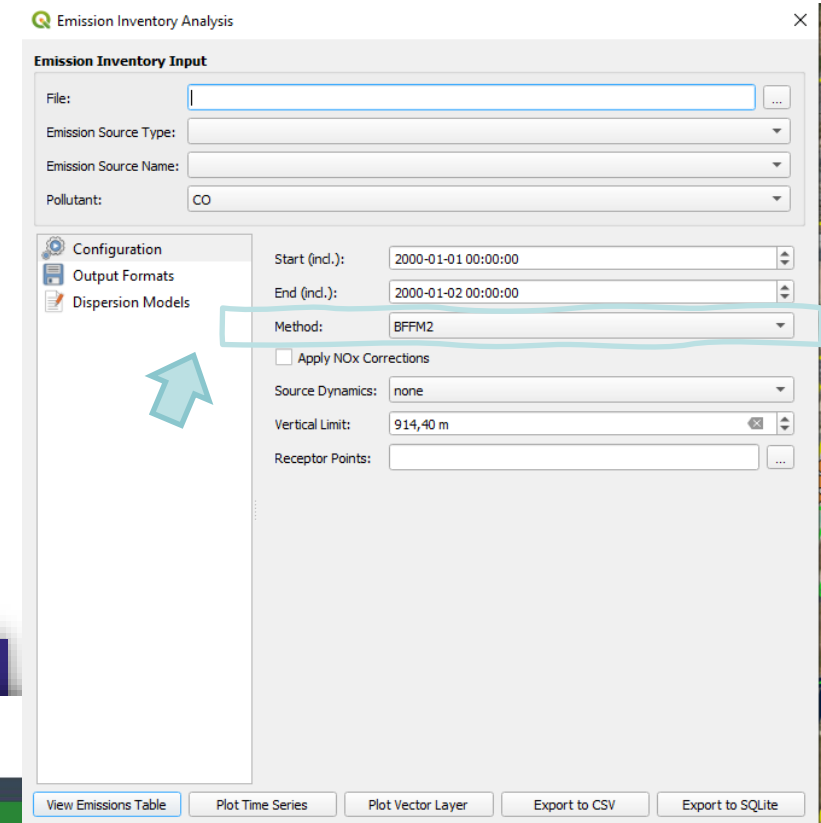
- ❑ By ICAO mode (TX, TO, CL, AP)
- ❑ Boeing Fuel Flow Method (BFFM2)

❑ Emission Factors:

- ❑ ICAO Engine Emissions Databank for aircraft jet engine
- ❑ FOCA Piston Engine EFs
- ❑ Possible to use **turboprops** EFs from the **FOI** data base

❑ Aircraft Performance:

- ❑ Aircraft Noise and Performance (ANP) Database



ICAO

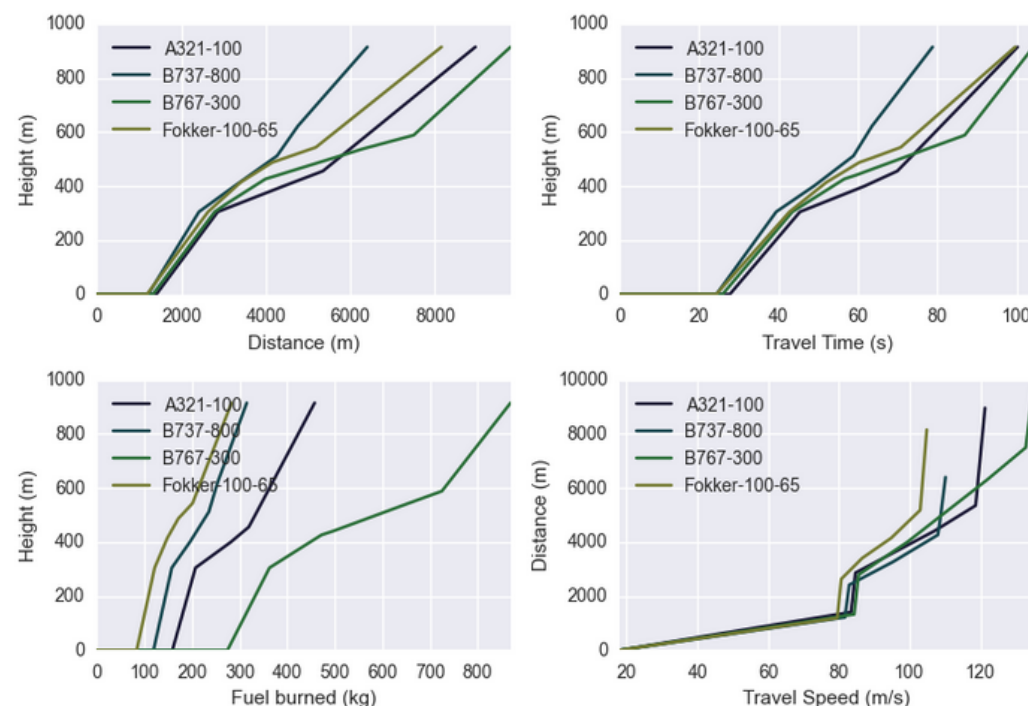
ENVIRONMENT

CAEP - Modelling and Databases Group
(MDG)



Aircraft Noise and Performance (ANP) Data

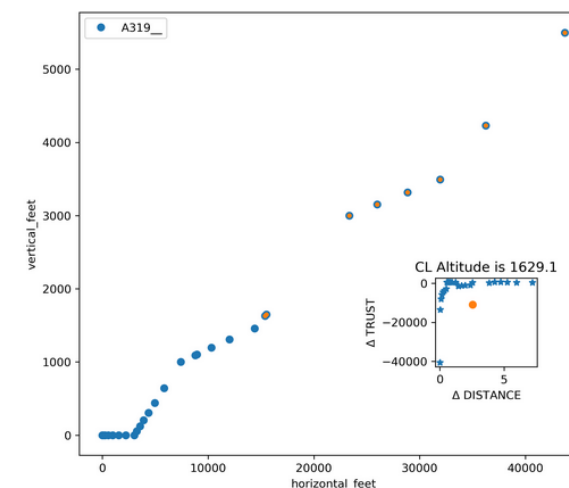
In OpenALAQS, the ANP fixed-point profiles (see [default aircraft profiles](#)) are used to calculate aircraft emissions. The ANP fixed-point profiles contain information on the relative 2D trajectory of the aircraft (horizontal versus vertical distance) on the runway. These points are converted into an aircraft trajectory for a given runway based on its geographic coordinates. Currently, only straight-line trajectories are possible in OpenALAQS.



Performance profiles

The ratio of thrust to distance is used to define the cut-off between take-off and climb-out. During take-off, full thrust is required to accelerate the aircraft. As the aircraft reaches a certain distance and speed, thrust is reduced to a level appropriate for climb. This transition involves reducing thrust from maximum take-off to maximum climb thrust after a set distance, typically around 1000 feet of ground distance. This cut-off point is used in OpenALAQS to separate the two modes.

The following figure illustrates this approach. For more information the user is referred to [ECAC.CEAC Doc 29, Volume 2, Appendix B](#).



Aircraft Noise and Performance (ANP) Data

<https://www.easa.europa.eu/en/domains/environment/policy-support-and-research/aircraft-noise-and-performance-anp-data>

Aircraft Noise and Performance (ANP) Data

DB Browser for SQLite - C:\Users\ssstromat\Dropbox\WorkOn\PROJECTS_ONLINE\OpenALAQS\open_alas_example\LSZH\LSZH.alas

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragma Execute SQL

Table: default_aircraft_profiles

	oid	profile_id	arrival_departure	stage	point	weight_lbs	horizontal_feet	vertical_feet	tas_knots	weight_kgs	horizontal_metre	vertical_metres	tas_metres	power	mode	course
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
7537	4326	JET-LARGE-A-1	A	1	3	369200	-68552.2	3000	216.2	167466.1664	-20894.71056	914.4	111.2227928	0.009915471	AP	ANP2.2
7538	4327	JET-LARGE-A-1	A	1	4	369200	-57243.4	3000	181.4	167466.1664	-17447.78832	914.4	93.3201416	0.01605654	AP	ANP2.2
7539	4328	JET-LARGE-A-1	A	1	5	369200	-48027.2	2517	170.4	167466.1664	-14638.69056	767.1816	87.6612576	0.017129677	AP	ANP2.2
7540	4329	JET-LARGE-A-1	A	1	6	369200	-46386.2	2431	166.7	167466.1664	-14138.51376	740.9688	85.7578148	0.017620113	AP	ANP2.2
7541	4330	JET-LARGE-A-1	A	1	7	369200	-40318.4	2113	150.5	167466.1664	-12289.04832	644.0424	77.423822	0.019967651	AP	ANP2.2
7542	4331	JET-LARGE-A-1	A	1	8	369200	-37979.2	1990.4	142.3	167466.1664	-11576.06016	606.67392	73.2053812	0.02121294	AP	ANP2.2
7543	4332	JET-LARGE-A-1	A	1	9	369200	-36979.2	1938	138.7	167466.1664	-11271.26016	590.7024	71.3533828	0.143731927	AP	ANP2.2
7544	4333	JET-LARGE-A-1	A	1	10	369200	-954.1	50	134.9	167466.1664	-290.80968	15.24	69.3984956	0.134183966	AP	ANP2.2
7552	3871	JET-LARGE-D-1	D	1	4	243300	9518.4	1721.7	157.9	110358.9336	2901.20832	524.77416	81.2307076	0.694292336	CL	ANP2.2
7559	3876	JET-LARGE-D-1	D	1	11	243300	91202.3	16000	205.3	110358.9336	10072.35192	3616	116.0200192	0.100730467	CL	ANP2.2
7560	3879	JET-LARGE-D-2	D	2	1	253000	0	0	0	114758.776	0	0	0	0.946236449	TO	ANP2.2
7561	3880	JET-LARGE-D-2	D	2	2	253000	3640.3	0	157	114758.776	1109.56344	0	80.767708	0.795365607	TO	ANP2.2
7562	3881	JET-LARGE-D-2	D	2	3	253000	9115.2	1500	160.5	114758.776	2778.31296	457.2	82.568262	0.830212897	TO	ANP2.2

7537 - 7563 of 8809

Go to: 1

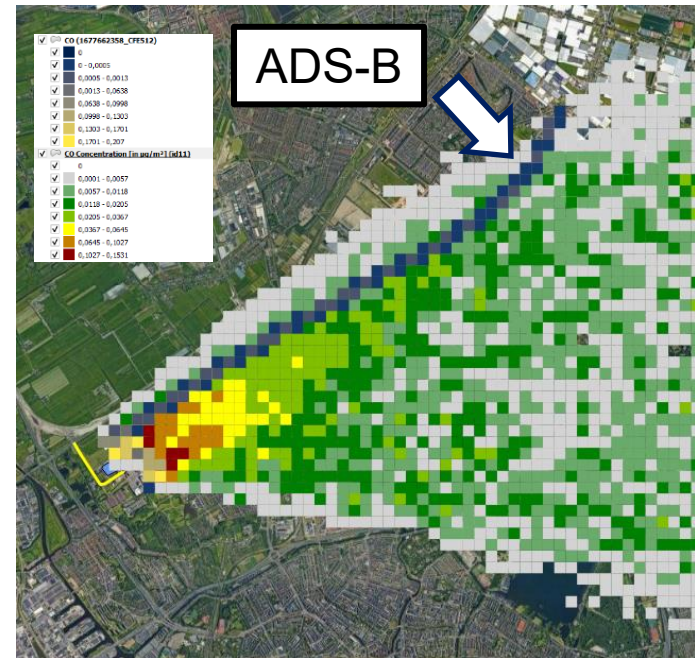
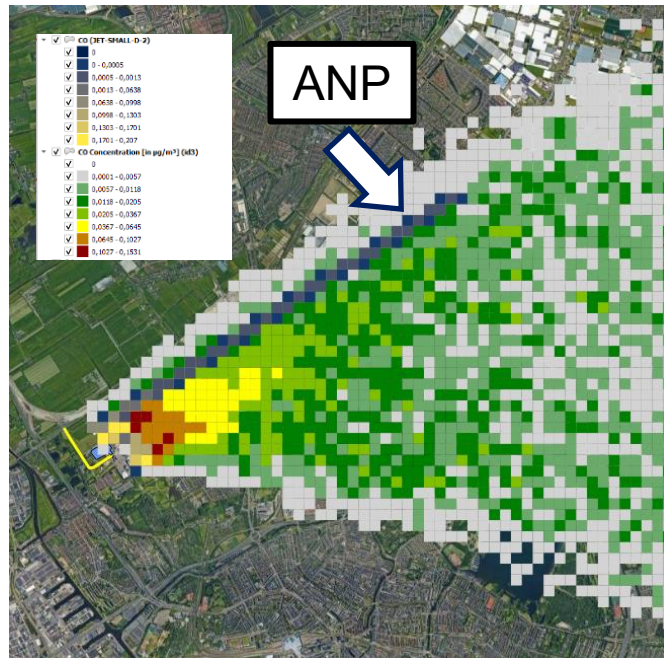
UTF-8

*Compare the emissions of the default profile for a single movement **vs** a profile of your choice*

Make the profile of your choice the default profile for the AC type of the same movement

Aircraft emissions: ANP vs ADS-B

- Possible to consider real aircraft trajectories (i.e. ADS-B data)
- Custom profiles must be provided by the user (same format as default aircraft profiles)

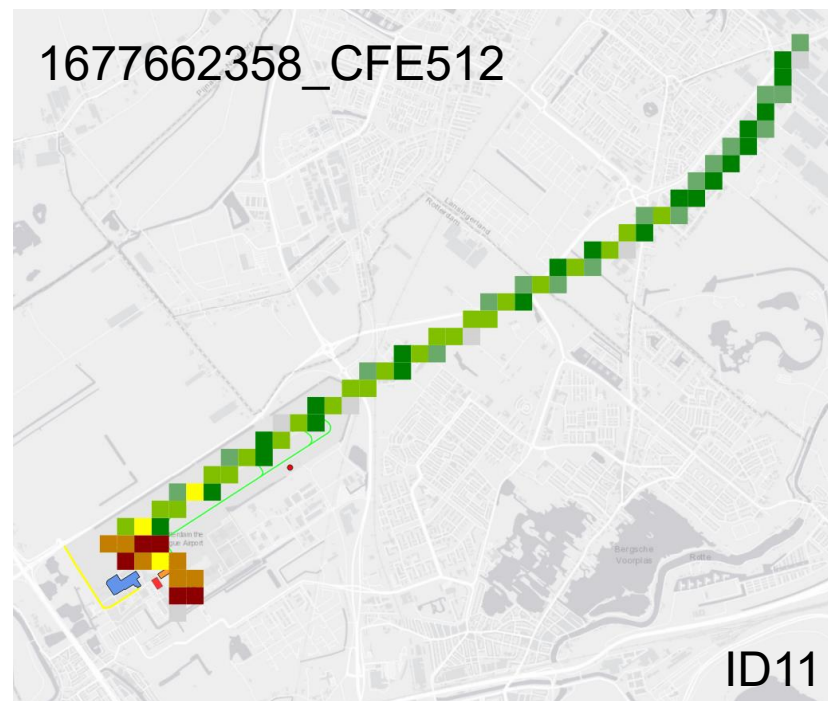
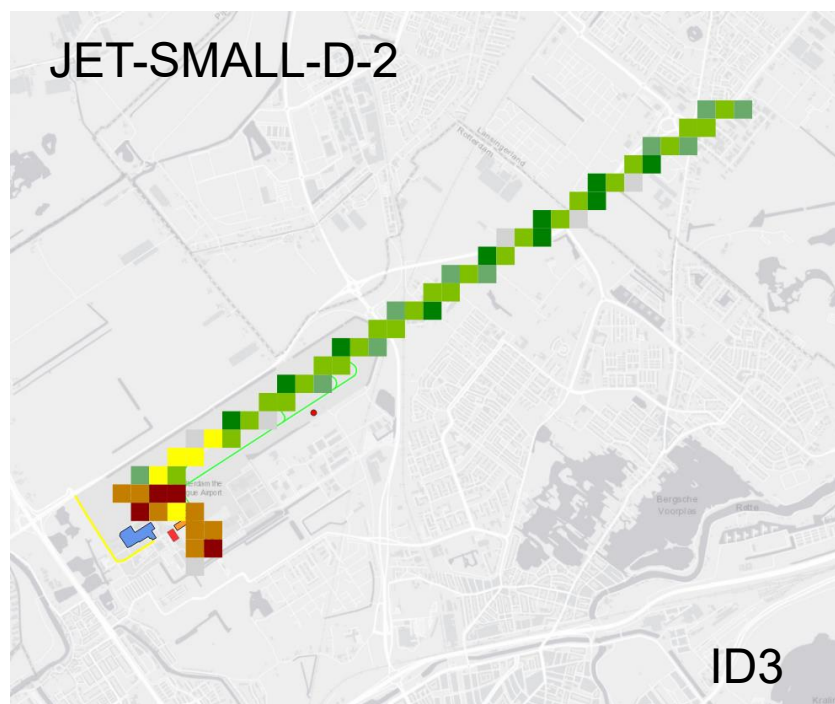


Aircraft Noise and Performance (ANP) Data



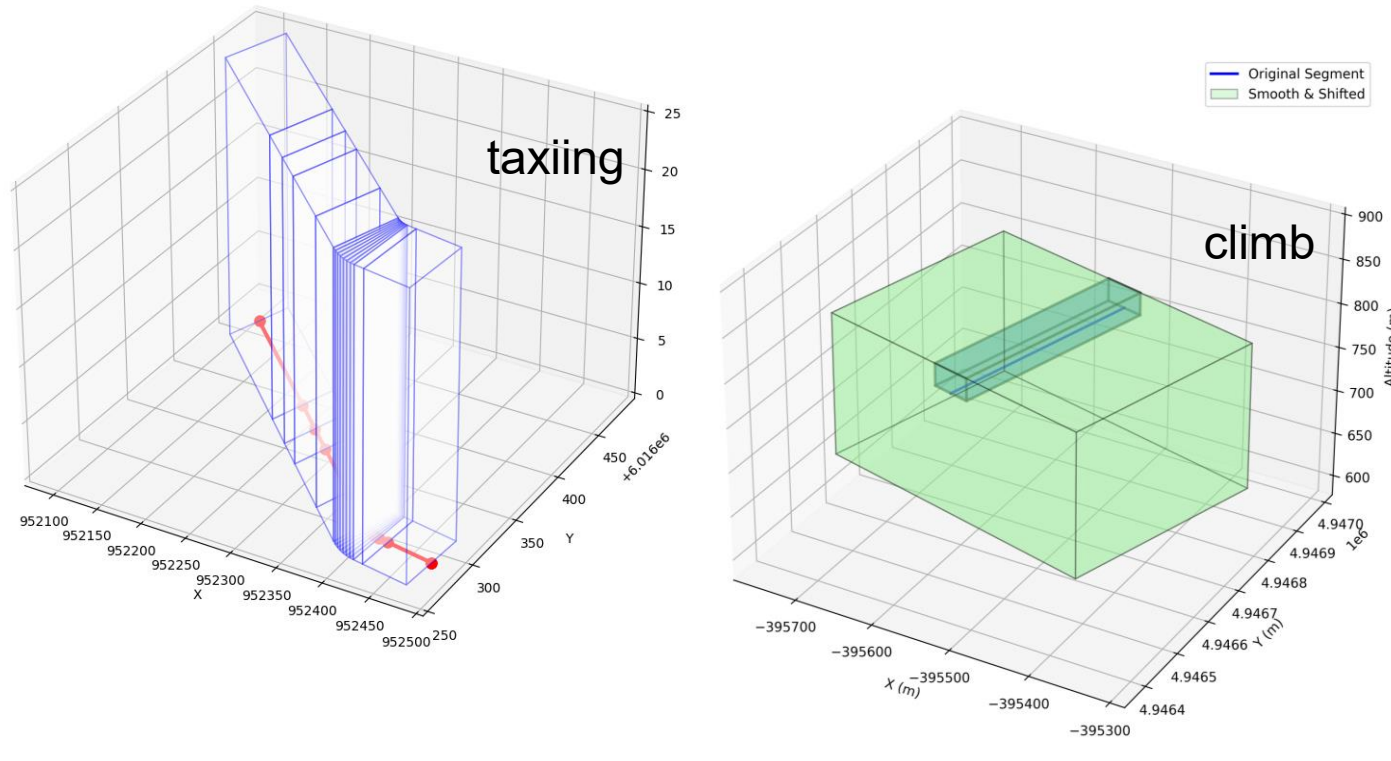
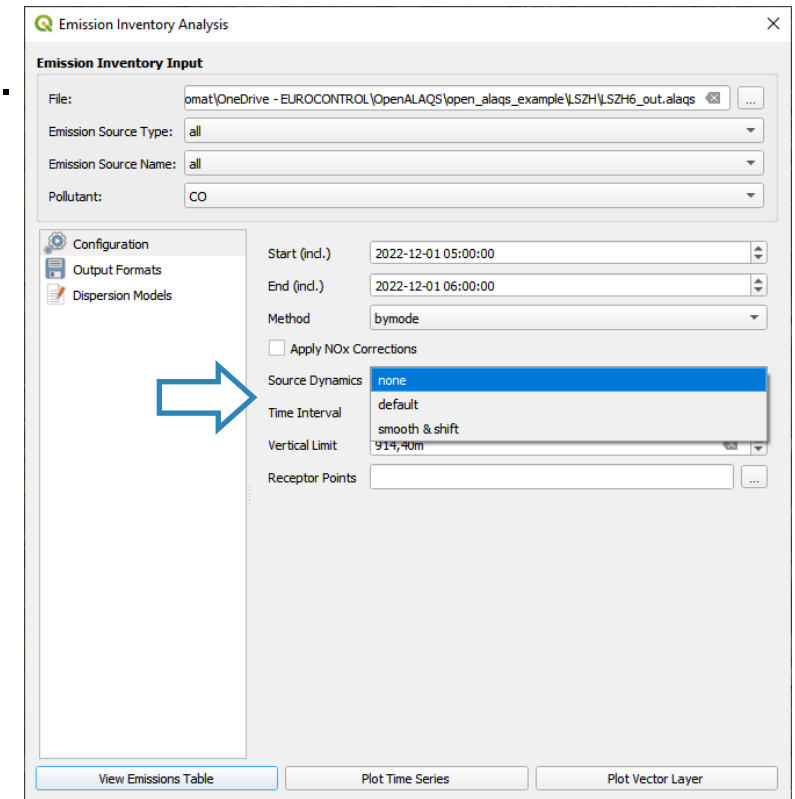
*Compare the emissions of the default profile for a single movement **vs** a profile of your choice*

3	3	2025-12-01 06:40:00	2025-12-01 06:35:00	A21N	G2	D	06	01P18PW157	JET-SMALL-D-2	G2/24/D/1
4	11	2025-12-01 06:41:00	2025-12-01 06:36:00	A21N	G2	D	06	01P18PW157	1677662358_CFE512	G2/24/D/1



Exhaust Dynamics (Smooth and Shift)

Possible to account for source dynamics such as turbulence, exhaust momentum from aircraft engines, and thermal plume rise.

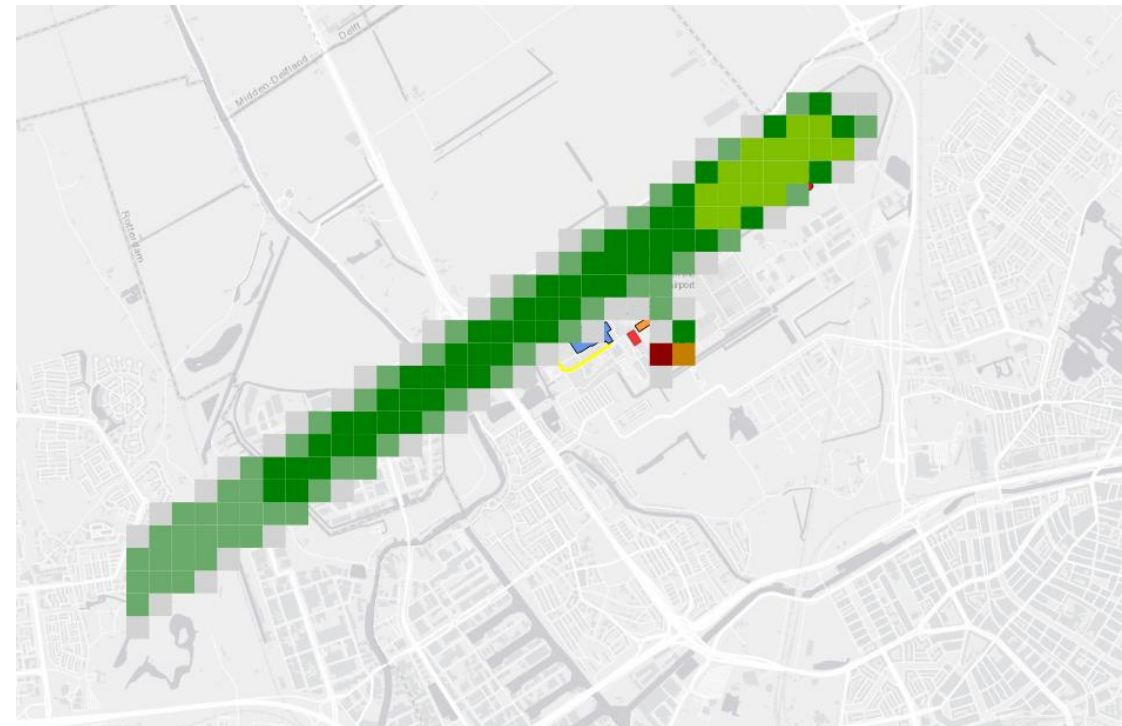



The screenshot shows the 'Emission Inventory Analysis' software interface. The 'Emission Inventory Input' section includes fields for File, Emission Source Type, Emission Source Name, and Pollutant. The 'Configuration' section includes fields for Start (incl.), End (incl.), Method, Apply NOx Corrections, Source Dynamics, Time Interval, Vertical Limit, and Receptor Points. A blue arrow points to the 'Source Dynamics' dropdown menu, which is open, showing options: none, default, smooth & shift, and 919,90m.

Exhaust Dynamics (Smooth and Shift)



Visualise the emissions of a single movement using the available options



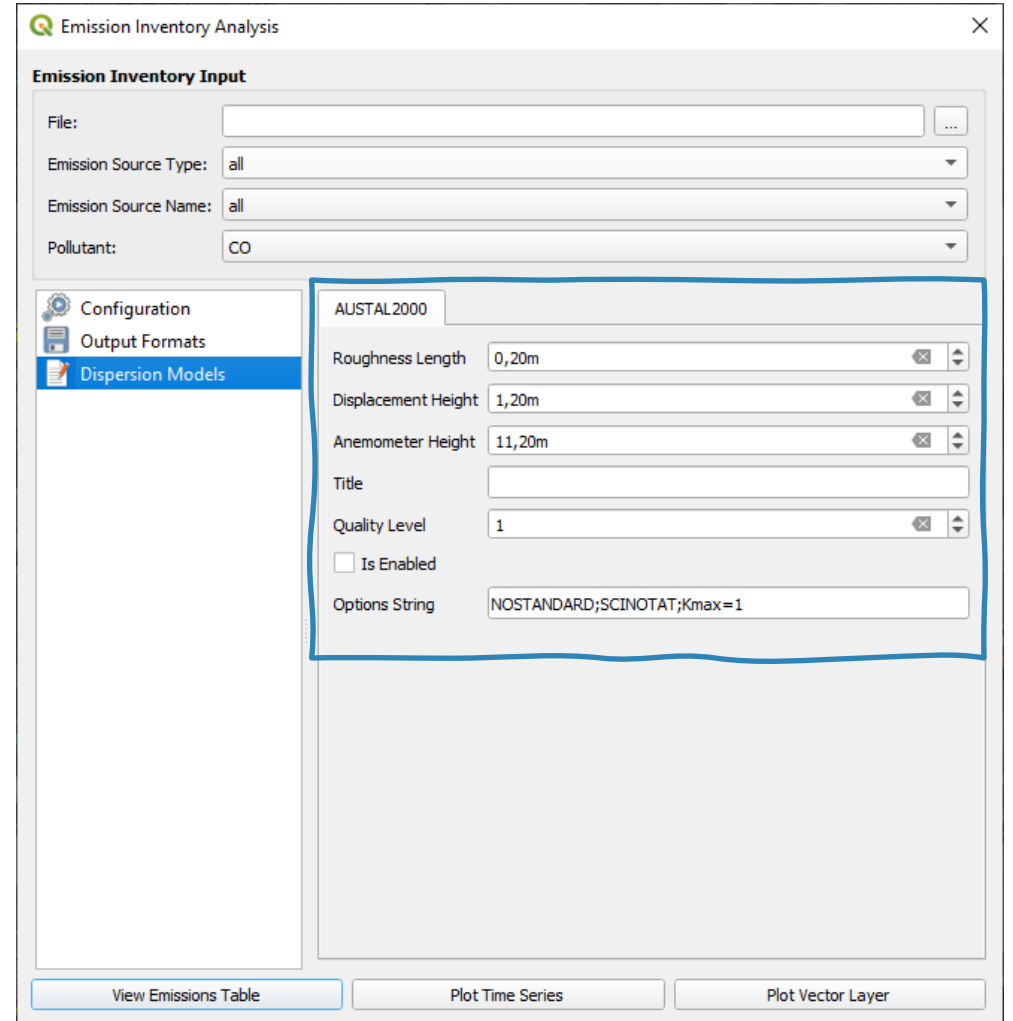
Dispersion modelling **AUSTAL**

- ❑ **Lagrangian particle model** in conformance with the German guideline VDI 3945 Part 3.
- ❑ The official reference model of the German Regulation on Air Quality Control (*Technische Anleitung zur Reinhaltung der Luft*, TA Luft)
- ❑ It **accounts for several different pollutants**, each with specific results, which are strictly bound to the regulations of the German Standard TA Luft 2002 (acc. to European EC Directive 2008/50/EC)
- ❑ AUSTAL (v3.3.0) is **provided free of charge** under the GNU Public Licence

<https://www.umweltbundesamt.de/en/topics/air/air-quality-control-in-europe/download>

AUSTAL Parameters:

- ☐ **Roughness Length** (*set to 0.2m by default*)
- ☐ **Displacement Height** (*set to 1.2m by default*)
- ☐ **Anemometer Height** (*set to 11.2m by default*)
- ☐ **Title**: The project name
- ☐ **Quality Level** (*set to 1 by default*)
- ☐ **Is Enabled**: Enable or disable dispersion module
- ☐ **Options String**: Define advanced AUSTAL settings



The screenshot shows the 'Emission Inventory Analysis' software window. The 'Emission Inventory Input' section at the top includes fields for 'File:', 'Emission Source Type:' (set to 'all'), 'Emission Source Name:' (set to 'all'), and 'Pollutant:' (set to 'CO'). Below this is a sidebar with three options: 'Configuration', 'Output Formats', and 'Dispersion Models', with 'Dispersion Models' currently selected. The main panel displays the 'AUSTAL2000' configuration settings, which are highlighted with a blue border. These settings include: 'Roughness Length' (0,20m), 'Displacement Height' (1,20m), 'Anemometer Height' (11,20m), 'Title' (empty), 'Quality Level' (1), 'Is Enabled' (unchecked), and 'Options String' (NOSTANDARD;SCINOTAT;Kmax=1). At the bottom of the window are three buttons: 'View Emissions Table', 'Plot Time Series', and 'Plot Vector Layer'.

https://github.com/eurocontrol-asu/open_alaqs/blob/main/open_alaqs/assets/emissions-calculation-2.png#dispersion-modeling-with-austal

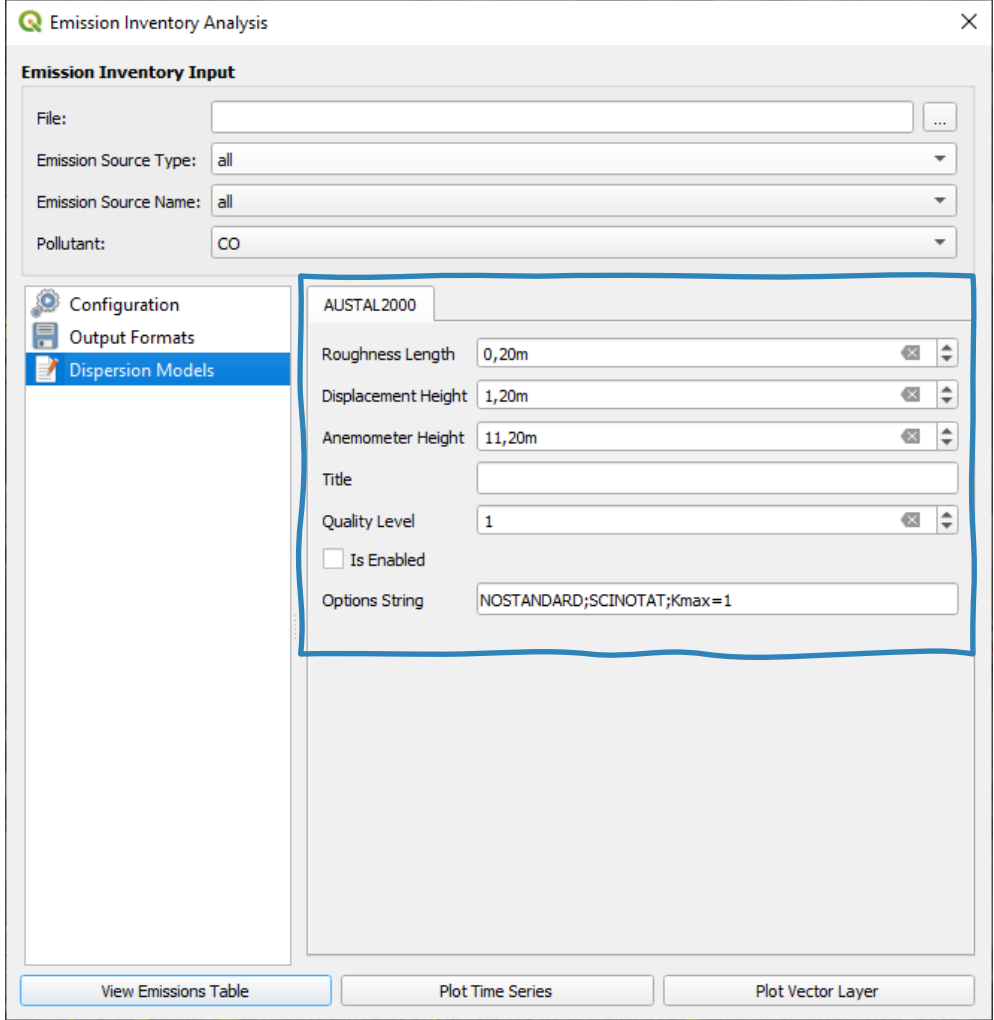
AUSTAL Parameters:

- ☐ **Roughness Length** (*set to 0.2m by default*)
- ☐ **Displacement Height** (*set to 1.2m by default*)
- ☐ **Anemometer Height** (*set to 11.2m by default*)
- ☐ **Title**: The project name
- ☐ **Quality Level** (*set to 1 by default*)
- ☐ **Is Enabled**: Enable or disable dispersion module
- ☐ **Options String**: Define advanced AUSTAL settings



Enable the dispersion module

Calculate emissions from all sources and prepare the AUSTAL input files



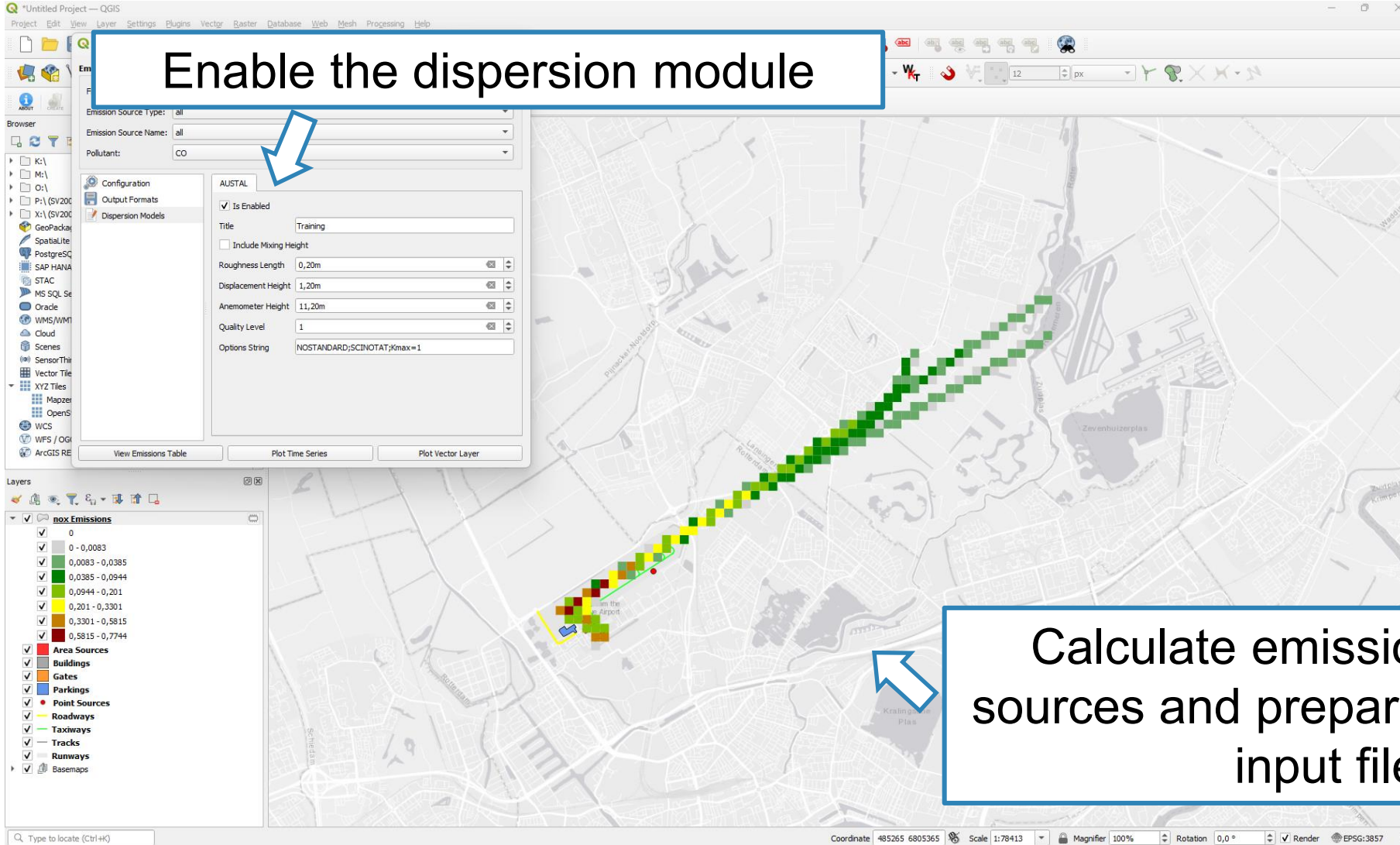
The screenshot shows the 'Emission Inventory Analysis' window. The 'Emission Inventory Input' section at the top includes fields for 'File:', 'Emission Source Type:' (set to 'all'), 'Emission Source Name:' (set to 'all'), and 'Pollutant:' (set to 'CO'). Below this is a sidebar with 'Configuration', 'Output Formats', and 'Dispersion Models' (which is selected). The 'AUSTAL2000' configuration panel is open, showing the following settings: 'Roughness Length' (0,20m), 'Displacement Height' (1,20m), 'Anemometer Height' (11,20m), 'Title' (empty), 'Quality Level' (1), 'Is Enabled' (unchecked), and 'Options String' (NOSTANDARD;SCINOTAT;Kmax=1). At the bottom of the window are three buttons: 'View Emissions Table', 'Plot Time Series', and 'Plot Vector Layer'.

AUSTAL Parameters:



Enable the dispersion module

Calculate emissions from all sources and prepare the AUSTAL input files



Running AUSTAL



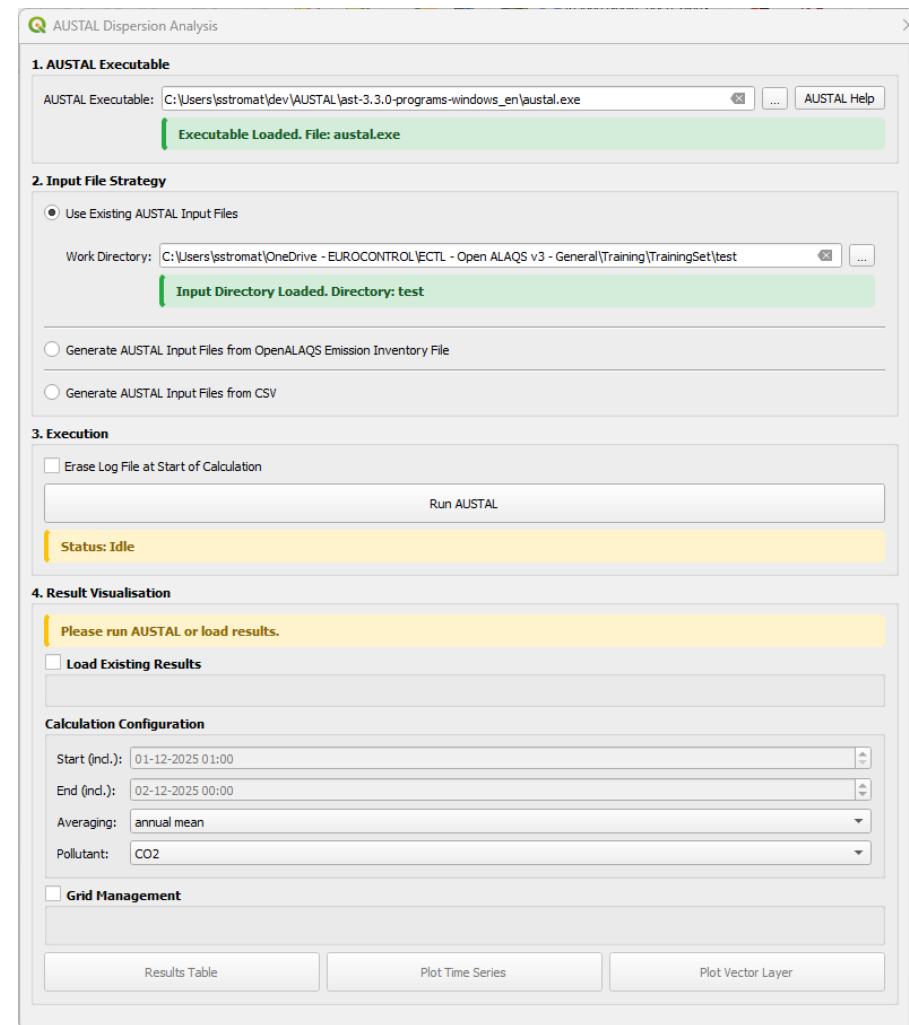
Specify the path to the **AUSTAL executable** and the project directory (**Work Directory**) where all output files will be written.

Run AUSTAL to start the dispersion calculation.

Select the .alaqs file used for generating the AUSTAL input data

Select the pollutant used and the averaging interval

Visualized the calculated concentrations on a map



The screenshot shows the AUSTAL Dispersion Analysis software interface. It is divided into four main sections:

- 1. AUSTAL Executable:** The AUSTAL Executable path is set to `C:\Users\jstromat\dev\AUSTAL\ast-3.3.0-programs-windows_en\ austal.exe`. A green status bar indicates "Executable Loaded. File: austal.exe".
- 2. Input File Strategy:** The "Use Existing AUSTAL Input Files" option is selected. The Work Directory is set to `C:\Users\jstromat\OneDrive - EUROCONTROL\ECTL - Open ALAQS v3 - General\Training\TrainingSet\test`. A green status bar indicates "Input Directory Loaded. Directory: test". Other options include "Generate AUSTAL Input Files from OpenALAQS Emission Inventory File" and "Generate AUSTAL Input Files from CSV".
- 3. Execution:** The "Erase Log File at Start of Calculation" checkbox is unchecked. A "Run AUSTAL" button is present. A yellow status bar indicates "Status: Idle".
- 4. Result Visualisation:** A yellow status bar indicates "Please run AUSTAL or load results." Below this, the "Load Existing Results" checkbox is unchecked. The "Calculation Configuration" section shows: Start (Ind.): `01-12-2025 01:00`, End (Ind.): `02-12-2025 00:00`, Averaging: `annual mean`, and Pollutant: `CO2`. The "Grid Management" checkbox is unchecked. At the bottom, there are buttons for "Results Table", "Plot Time Series", and "Plot Vector Layer".

Running AUSTAL



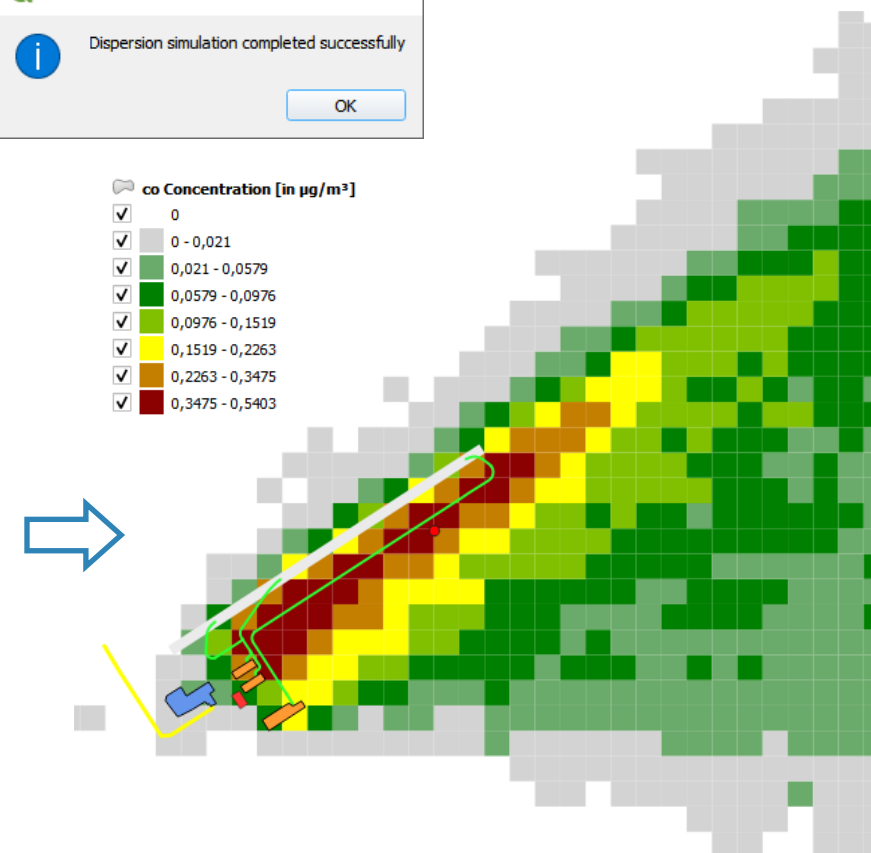
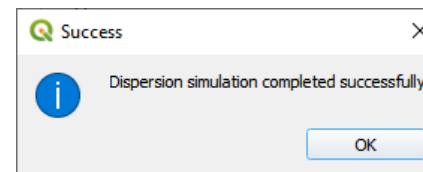
Specify the path to the **AUSTAL executable** and the project directory (**Work Directory**) where all output files will be written.

Run AUSTAL to start the dispersion calculation.

Select the .alaqs file used for generating the AUSTAL input data

Select the pollutant used and the averaging interval

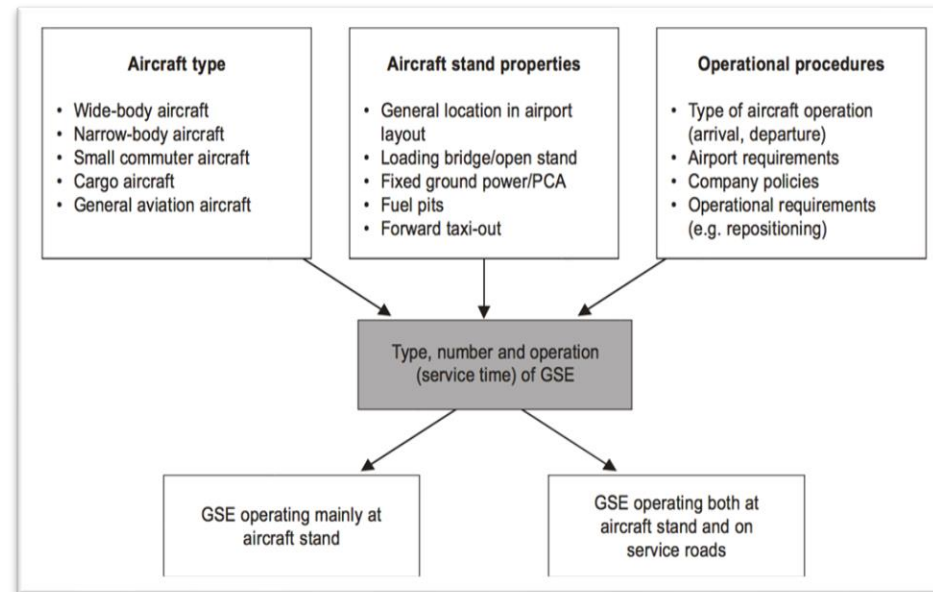
Visualized the calculated concentrations on a map





GSE custom emissions application

- Standalone tool to prepare custom EFs for GSE
- Default values provided – easily editable
- Methodology follows ICAO's Airport Air Quality Manual (Doc. 9889)



ICAO's Airport Air Quality Manual (Doc. 9889)

GSE Application

- Import movements from .alaqs db
- Select GSE per AC/Gate/Operation type – Number and time
- Calculate custom EFs and export/update .alaqs db

GSE Emissions Calculator

Open Database Assign GSE Calculate Emissions Exit Application

GSE Emission Standards Movements

type	description	power	load	fuel	Stage	time	deterioration
GPU	400 Hz Groun...	142.0	0.5	Diesel	Stage I	1.0	1.0
GSE	Baggage Belt ...	33.0	0.25	Petrol	Stage I	1.0	1.0
GSE	Baggage Cart ...	0.0	0.0	Electric	Stage I	1.0	1.0
GSE	Cabin Cleanin...	132.0	0.1	Diesel	Stage I	1.0	1.0
GSE	Cabin Cleanin...	61.0	0.1	Petrol	Stage I	1.0	1.0
GSE	Cabin Cleanin...	48.0	0.1	Petrol	Stage I	1.0	1.0
GSE	Cargo Delivery	33.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cargo Loader	62.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cargo Loader...	59.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cars BZRA/B	50.0	0.2	Petrol	Stage I	1.0	1.0
GSE	Catering Truck	110.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Cleaning ...	3.5	0.8	Petrol	Stage I	1.0	1.0
GSE	Cleaning ...	2.2	0.8	Petrol	Stage I	1.0	1.0
GSE	Fork Lift	66.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Gear mounte...	60.0	0.25	Diesel	Stage I	1.0	1.0
GSE	GSE/GA Fuel ...	90.0	0.1	Diesel	Stage I	1.0	1.0
GSE	Large Fork Lift	75.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Lavatory Truck	110.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Line ...	95.0	0.25	Diesel	Stage I	1.0	1.0
GSE	Mobile Air ...	150.0	0.5	Diesel	Stage I	1.0	1.0

Add Row Remove Row Save Changes

Assign GSE

JET SMALL/PIER/D

Assigned	type	description	power	load	fuel	Stage	Time	Count
☒	GPU	400 Hz Ground Power Unit	142.0	0.5	Diesel	Stage I	10.0	1
☒	GSE	Baggage Belt loader	33.0	0.25	Petrol	Stage I	10.0	1
☐	GSE	Baggage Cart Tractor	0.0	0.0	Electric	Stage I	10.0	1
☐	GSE	Cabin Cleaning Truck	132.0	0.1	Diesel	Stage I	10.0	1
☐	GSE	Cabin Cleaning Van	61.0	0.1	Petrol	Stage I	10.0	1
☐	GSE	Cabin Cleaning Vehicle	48.0	0.1	Petrol	Stage I	10.0	1
☐	GSE	Cargo Delivery	33.0	0.25	Diesel	Stage I	10.0	1
☐	GSE	Cargo Loader	62.0	0.25	Diesel	Stage I	10.0	1
☐	GSE	Cargo Loader large (B747 Main Deck)	59.0	0.25	Diesel	Stage I	10.0	1
☐	GSE	Cars BZRA/B	50.0	0.2	Petrol	Stage I	10.0	1
☐	GSE	Catering Truck	110.0	0.25	Diesel	Stage I	10.0	1

Summary:

Aircraft Group	Gate Type	Arr/Dep	GSE Type	Description	Power	Time	Count	Deterioration
1 JET SMALL	PIER	D	GPU	400 Hz Ground Power Unit	142.0	10.0	1	1.0
2 JET SMALL	PIER	D	GSE	Baggage Belt loader	33.0	10.0	1	1.0

OK Cancel

Emissions Calculator

GSE List by Movement

movement_code	type	description	power	load	fuel	time	count	deterioration_factor
1 JET SMALL/PIER/D	GPU	400 Hz Ground Power Unit	142.0	0.5	Diesel	10.0	1	1.0
2 JET SMALL/PIER/D	GSE	Baggage Belt loader	33.0	0.25	Petrol	10.0	1	1.0

Calculate Emissions

Output Emissions

oid	gate_type	ac_group	emis_type	time_unit	op_type	time	emis_unit	co	hc	nox	sox	pm10	source
1 1	PIER	JET SMALL	GSE	minutes	A	0.0	grams/hour	8.94	1.79	12.65	0.0	1.17	GSE Application
2 2	PIER	JET SMALL	GSE	minutes	D	10.0	grams/hour	8.94	1.79	12.65	0.0	1.17	GSE Application
3 3	PIER	JET SMALL	GPU	minutes	A	0.0	grams/hour	59.17	15.38	108.87	0.0	6.39	GSE Application
4 4	PIER	JET SMALL	GPU	minutes	D	10.0	grams/hour	59.17	15.38	108.87	0.0	6.39	GSE Application

Export table to CSV
Update the OpenALAQS database

https://github.com/eurocontrol-asu/open_alaqs/tree/main/gse_application

Customising OpenALAQs

Development

[\(Back to top\)](#)

1. Clone this repository.
2. Optionally, create a soft link between the local checkout and the QGIS plugin directory for easier developement. (Linux instructions: `ln -s ${PWD} ${HOME}/.local/share/QGIS/QGIS3/profiles/default/python/plugins/open_alaq/`).
3. Install [pre-commit](#) .
4. Develop a new feature.
5. Open a PR.
6. Wait for the CI to succeed.
7. Ensure you have a PR approval from another reviewer.
8. Merge the PR.

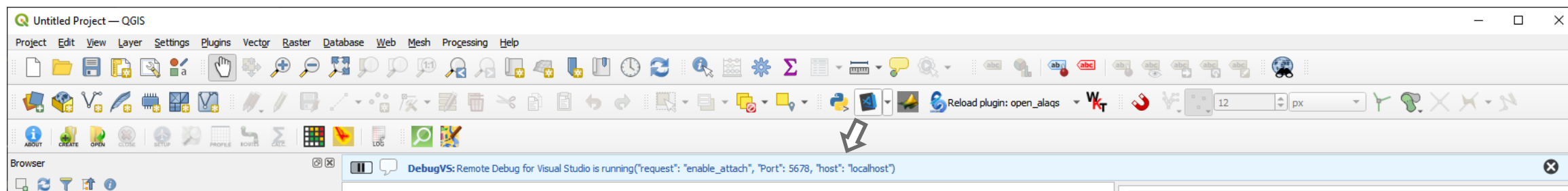
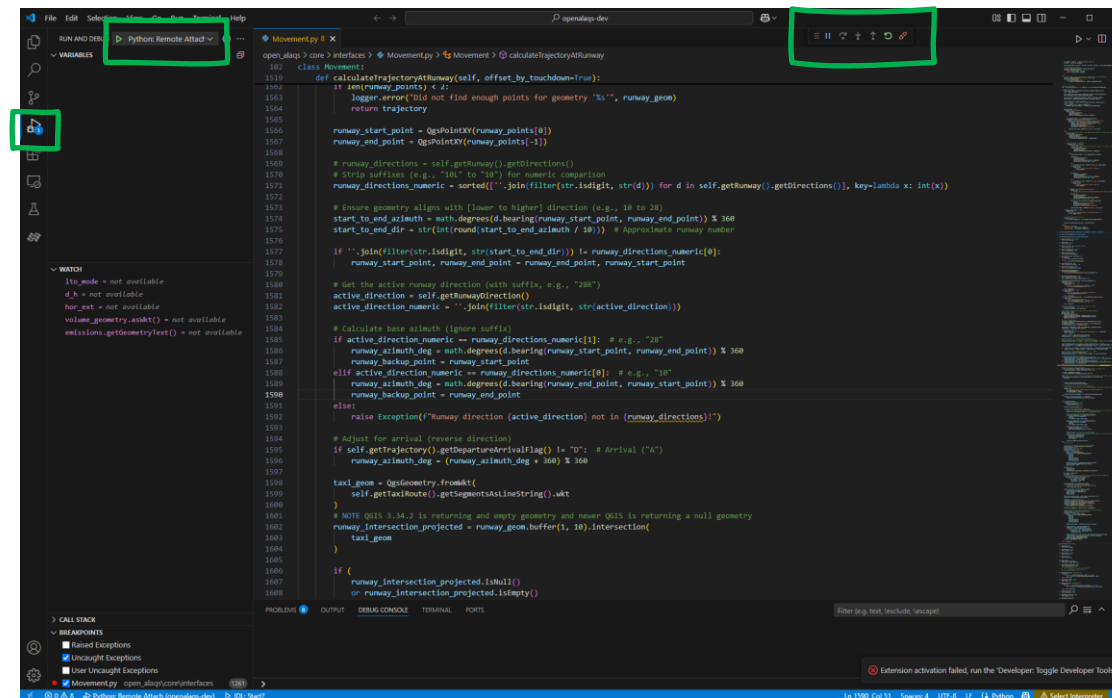
Customising OpenALAQS

Debugging

Debugging can be done via [QGIS VSCode Debug plugin](#) and [VSCode](#).

Sample `launch.json`:

```
{
  // Use IntelliSense to learn about possible attributes.
  // Hover to view descriptions of existing attributes.
  // For more information, visit: https://go.microsoft.com/fwlink/?linkid=830387
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Python: Remote Attach",
      "type": "python",
      "request": "attach",
      "port": 5678,
      "host": "localhost",
      "pathMappings": [
        {
          "localRoot": "${workspaceFolder}/open_alajs",
          "remoteRoot": "${HOME}/.local/share/QGIS/QGIS3/profiles/default/python/plugins/open_alajs/"
        }
      ]
    }
  ]
}
```



SUPPORTING EUROPEAN AVIATION

